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Reconstructing North Atlantic Ocean Heat Content Using Convolutional Neural Networks

Master Thesis

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Abstract

Slowly varying ocean heat content is one of the most important variables when describing climate variability on interannual to decadal time scales. Since observation-based estimates of ocean heat content require extensive observational coverage, incomplete observations are often combined with numerical models via data assimilation to simulate the evolution of oceanic heat. However, incomplete observations, particularly in the subsurface ocean, lead to large uncertainties in the resulting model-based estimate. As an alternative approach, Kadow et al., 2020 have proven that artificial intelligence can successfully be utilized to reconstruct missing climate information for surface temperatures. In the following, I investigate the possibility to train their three-dimensional convolutional neural network to reconstruct missing subsurface temperatures to obtain ocean heat content estimates with a focus on the North Atlantic ocean.

The network is trained and tested to reconstruct a 16 member Ensemble Kalman Filter assimilation ensemble constructed with the Max-Planck Institute Earth System Model for the period from 1958 to 2020. Specifically, I examine whether the partial convolutional U-net represents a valid alternative to the Ensemble Kalman Filter assimilation to estimate North Atlantic subpolar gyre ocean heat content. Additionally, I draw conclusions from this reconstruction in regard to the assimilation's uncertainty outside of its ensemble spread.

The neural network is capable of reproducing the assimilation reduced to datapoints with observational coverages within its ensemble spread with a correlation coefficient of 0.93 over the entire time period and of 0.99 over 2004 – 2020 (the Argo-Era). Additionally, the network is able to reconstruct the observed ocean heat content directly from observations for 12 additional months with a correlation of 0.97, essentially replacing the assimilation experiment by an extrapolation. When reconstructing the pre-Argo-Era (1958 – 2004), the network's training data still consists only of data assimilations from the Argo-Era, which are assumed to possess lower uncertainties due to more observational input. The lower correlation in the resulting reconstruction of masked assimilations from the pre-Argo-Era indicates significant uncertainties in the assimilation outside of its ensemble spread at times with low observational density. My results therefore demonstrate that the neural network is not only capable of reproducing the observed ocean heat content over its training period, but also before and after as well as improving estimates of the assimilation's uncertainty making it a suitable candidate to step-wise extend or replace data assimilation.

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1. Introduction

Predicting changes of Earth system variables on interannual to decadal scales has been the focus of large amounts of current climate research. From forecasts of extreme weather events like droughts or heat waves to predictions of precipitation and temperature on regional scales, decadal predictions are the center of a lot of scientific debate (Keenlyside and Ba, 2010; Marotzke et al., 2016). Land surface and atmospheric components of the Earth system often change so quickly that they rarely possess any memory beyond seasonal changes. Thus, predictions on larger timescales from months to decades cannot solely rely on them as indicators for future changes. Contrary to its land and atmospheric components, the global oceans encompass low frequency variability and a much longer memory. Therefore, they play an important role in climate modeling on interannual to decadal time scales. In particular ocean heat content (OHC) indicates long term changes in heat fluxes between the Earth system's oceanic and atmospheric components (Cheng et al., 2017). In addition to that, OHC also shows a consistent and slowly varying response to changes in external boundary conditions of the climate system. Therefore, representing the overall changes in ocean heat content as well as its variability and response to external forcings is one of the major challenges for decadal climate predictions.

Ocean heat content is not only important for climate predictions due to its low frequency variability. In comparison to surface land and sea temperatures, changes in ocean heat content also show lower background noise in comparison to their signal, making them a better indicator for climatic changes (Cheng et al., 2018). Additionally, OHC is instrumental to the Earth's energy budget and its response to anthropogenic greenhouse gas emissions. Greenhouse gases reflect longwave radiation back towards the Earth's surface, creating an imbalance in the Earth's energy budget in the form of a surplus of incoming energy compared to the outgoing longwave radiation. More than 90% of this energy imbalance is retained in the global oceans with over 30% being stored in the North Atlantic (Häkkinen et al., 2015). This does not only lead to thermal expansion and thus a direct rise in sea level but also affects the formation of sea ice and the global circulation systems. Contrary to most of the global oceans, not all of the North Atlantic OHC is increasing over the past century. Especially in the subpolar gyre (SPG), OHC has been overall declining over the past 70 years. Most of this change can be attributed to natural variability related to the North Atlantic Oscillation (NAO) (Lozier et al., 2008). This natural variability in the North Atlantic and its connection to Northern European and American regional climate is another reason, why the study of North Atlantic and SPG OHC has gotten a lot of attention in recent literature.

The North Atlantic and especially its sea surface temperature (SST) can be predicted with relatively high skill on interannual to decadal timescales (Buckley et al., 2019). However, especially when examining subsurface temperatures and OHC in the SPG, larger uncertainties remain. Better estimating these uncertainties could thus give insights into the different regional OHC trends in the North Atlantic and their impact on European climate and weather.

In general, ocean heat content estimates are dependent on several factors, which can all impact

the final result with their incorporated uncertainties. A first and very direct source of uncertainty is the bias measurement instruments bring into the calculation of ocean heat content. For example, over large parts of the 20th century, mechanical bathythermographs (MBTs) and expendable bathythermographs (XBTs) provided considerable amounts of ocean temperature profile data. However, both measurement methods have been found to incorporate a time variant warm bias, which significantly impacted ocean heat content timeseries. Even after more sophisticated bias corrections over the past decades, a bias of up to 0.1°C remains, particularly in the southern hemisphere (Gouretski and Cheng, 2020). In addition to direct measurements, Boyer et al., 2016 have identified mapping methods and baseline climatologies as additional sources of uncertainties. Especially since scientific consensus on the choice of XBT bias correction method has been reached by now (Garcia-Soto et al., 2021), Boyer et al., 2016 suggest that the choice of mapping method remains as the largest source of uncertainty for the calculation of ocean heat content.

Mapping methods refer to the way of reconstructing a full set of values of a variable from a limited amount of observations. As ocean heat content cannot be observed directly, it is usually calculated by integrating over temperature profiles from the ocean surface down to a depth of up to 2.000 m. Below 2.000 m, very few observations exist, as measurement instruments, such as Argo floats, do not descend below this threshold Wong et al., 2020. Thus, even sophisticated mapping methods have significant difficulties in estimating deeper ocean heat content. However, even above 2.000 m depth, subsurface temperature observations often lack adequate coverage and thus cannot be used without further processing to initialize climate models. In order to create a complete set of values close to the observed state, a range of methods can be applied. On the one hand, statistical methods interpolate between missing data points on a purely mathematical basis. Statistical interpolation approaches, such as kriging, are relatively cost effective and have shown reasonable success in reconstruction temperature data (Varentsov et al., 2020). However, most statistical infilling has shown difficulties to map variables in undersampled regions. This difficulty mostly stems from their relaxation back to the underlying baseline climatology without considering larger dynamical trends or physical processes. The other larger field, data assimilation, tries to counter these difficulties by aiming to bring together Earth system model states and real life observations. It assumes both to be inaccurate and incomplete representations of the state of a system. Data assimilation then tries to incorporate both into a new statistical estimation of the true state. Of the many data assimilation methods, this study will utilize on Ensemble Kalman Filter (EnKF) assimilations. In comparison to other data assimilations, EnKFs provide computationally efficient and very accurate results compared to other assimilation methods, especially in high dimensional systems, such as the climate and Earth system (Carrassi et al., 2018). A detailed overview of the mechanisms behind EnKFs as well as other assimilation methods can be found in Carrassi et al., 2018.
In general, statistical interpolation and data assimilations have shown high correlation to each other in the reconstruction of oceanic heat content Brune et al., 2018. However, in the 20th century, data assimilation runs show lower correlation to interpolation methods compared to the last 20 years (Brune et al., 2018). The biggest difference between both time periods for the

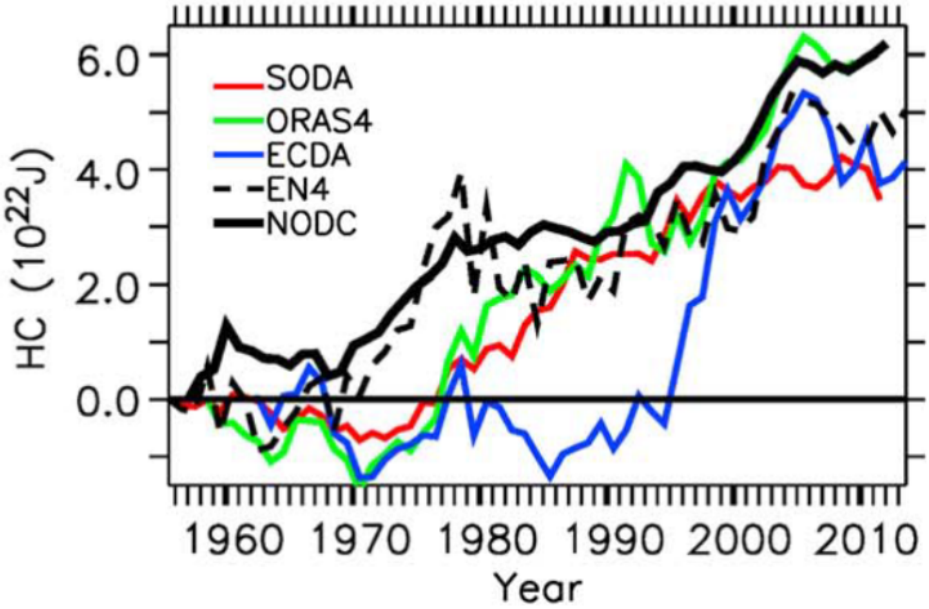

Figure 1.1.: *Estimates of upper 2.000 m North Atlantic ocean heat content using different data assimilation and interpolation methods (Häkkinen et al., 2015).*

mapping of OHC is the amount of subsurface temperature observations available in the reconstruction process. At the start of the 2000s the amount of subsurface temperature observations in global oceans started to rise rapidly due to the introduction of the Argo project. The Argo project placed 4.000 floats in the global oceans (Roemmich et al., 2009), which drift with ocean currents and continually measure temperatures up to a depth of 2.000 m. Due to their significant impact on the availability of NA subsurface temperature observations, the time before the introduction of Argo floats in 2004 will from hereon out be referred to as pre-Argo-Era, while the time from 2004 onwards will be referred to as Argo-Era. The timeseries of different infilling methods of global upper ocean heat content show higher correlation with each other during the Argo-Era, as can be seen in figure 1.1. Only for a short period during the 1960s do they agree with each other. This is probably due to all estimates more or less reverting back to their baseline climatologies at times with very few observations. As there is no reference point for the ground truth available at points without direct observation, the only estimate of the missing data comes in form of uncertainty ranges of different infilling methods. Therefore, the uncertainty range of ocean heat content estimates at locations with undersampled temperature measurements is generally higher during the pre-Argo-Era (Levitus et al., 2012).

This uncertainty is also visible when examining one specific assimilation ensemble. Figure 1.2 shows the OHC estimate of a 16 member EnKF assimilation ensemble (Brune and Baehr, 2020). The technical details of this assimilation are described in detail in section 2.1.1. The ensemble spread, which represents the uncertainty within the ensemble, is significantly larger during the pre-Argo-Era than during the 2000s. In addition to rising uncertainty with fewer observations, EnKF assimilations also require high computational resources to assimilate large datasets. Even though they are still comparatively efficient when compared to other assimilation methods, high resolutions and larger ensembles can significantly prolong the computation time. Thus, finding a

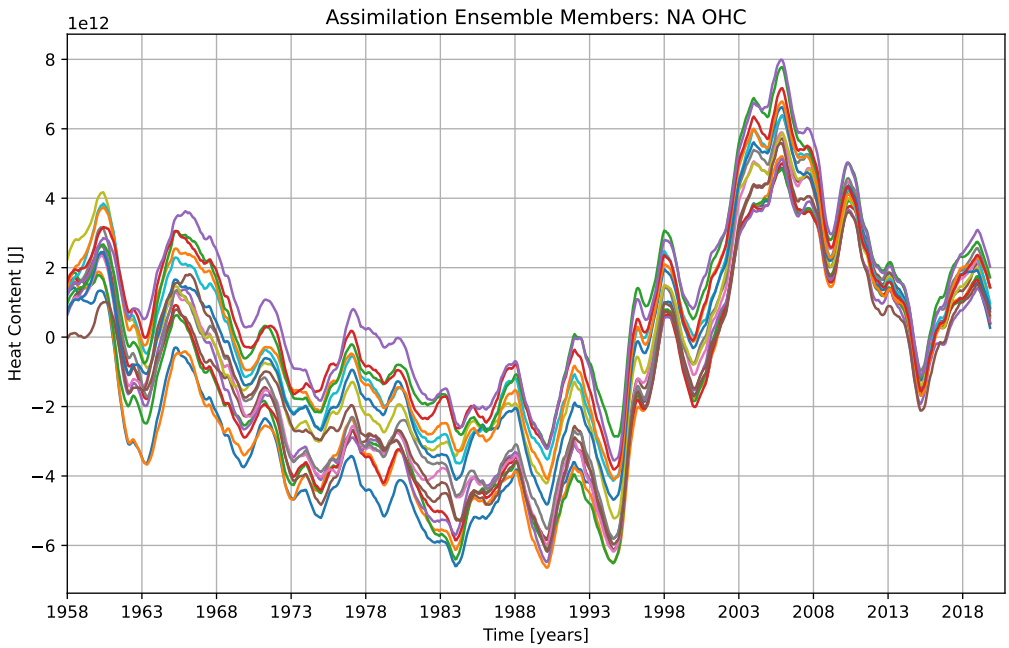

Figure 1.2.: *Ensemble spread of an Ensemble Kalman Filter (EnKF) data assimilation of North Atlantic OHC.*

more efficient alternative to assimilations similar to the utilized EnKF could potentially surpass its current computational limitations while also providing a more accurate uncertainty estimate in times and regions´ with low observational density.

In recent years, data assimilation techniques have often been combined with machine learning approaches, more specifically with neural networks, to improve the reconstruction of full system states. This combination aims at improving data assimilation methods (Brajard et al., 2021; Gottwald and Reich, 2021) with the assistance of neural networks as well as replacing data assimilation completely by machine learning (Geer, 2021). Machine learning has the advantage of combining numerical infilling with Earth system model (ESM) data (Kadow et al., 2020). Neural networks in particular, can incorporate the spatial awareness of ESMs, while keeping the computational efficiency of statistical infilling. Especially convolutional neural networks have proven capable of capturing spatial patterns of their training input. In 2020, Kadow et al., 2020 managed to transfer deep learning techniques originally developed for image inpainting to the field of climate sciences. They adapted the inpainting techniques developed by Liu et al., 2018 to reconstruct incomplete global 2m-air temperature data. The reconstructed Climate Model Intercomparison Project Phase 5 (CMIP5) data showed lower root mean squared errors (RMSE) than several kinds of statistical mapping methods. Furthermore, the neural network used by Kadow et al., 2020 was able to successfully reconstruct the historically well recorded 1877 El Nino event from observational data, which both statistical mapping methods did not reveal.
In comparison to traditional assimilations, machine learning approaches can reduce the computational costs significantly especially when working with large datasets (Arcucci et al., 2021).

Furthermore, machine learning approaches can integrate changes in observational data and new ensemble members much more easily as they are independent of an entire model setup. Careful selection of training data from periods with high observational density can lastly also limit model uncertainties to a minimum, as a trained network might be able to transfer trained patterns from periods with many observations to periods with few. Thus, it can learn the more accurate assimilation patterns from times with high observational density. Transferring these to earlier time periods, it is examined if the neural network can show inconsistencies in the assimilation caused by missing observations. Additionally, training a neural network on assimilation data and then reconstructing observations directly can help show inconsistencies and disagreements between observations and assimilation at the grid points with observations.

In order to utilize these advantages, I will build on the theoretical work of Arcucci et al., 2021 by training a neural network based on the one used by Kadow et al., 2020 to reconstruct three dimensional North Atlantic temperature data. This approach aims at answering two main questions surrounding the current estimates of SPG OHC. On the one hand, it aims at assessing the potential of neural networks to reproduce and even replace EnKF data assimilation of SPG OHC. Training a neural network on EnKF assimilations, its ability to reproduce assimilation estimates of SPG OHC will be tested. On the other hand, it will be ascertained if a neural network reconstruction of SPG OHC can help better estimate the uncertainty of EnKF assimilations, especially during the pre-Argo-Era.

In summary, this thesis will focus on examining whether neural networks provide a valid alternative to EnKF assimilations when mapping out SPG OHC, while at the same time trying to better estimate their uncertainties. In order to reach these aims, at first an overview of the involved data and methods will be given in chapter 2. Both the EnKF assimilation used as training data and the observations utilized in these assimilations and for the later reconstructions will be described in detail in section 2.1. Additionally, two other OHC estimates, the EN4 data set created by the Met Office Hadley Center and the Institute for Atmospheric Physic's (IAP) OHC dataset, will be reviewed to later provide some context to the reconstruction results. Afterwards, the utilized neural network will be described in section 2.2. Then, the results of the neural network reconstructions will be described, analyzed and interpreted in chapter 3. Finally, these results will be discussed in regard to their limitations and the underlying assumptions in chapter 4 before drawing final conclusions in chapter 5.

2. Data and Methods

2.1. Data

In order to gain a new estimate of North Atlantic ocean heat content, this work will use two different sources of training and testing data for the utilized neural network. On the one hand, data assimilation experiments (Brune and Baehr, 2020) from the Max Planck Institute Earth System Model (MPI-ESM, low resolution, Mauritsen et al., 2019) are deployed to train the neural network to reconstruct temperature maps. On the other hand, observational data from the fourth version of the Met Office Hadley Center 'EN' series (EN4, Good et al., 2013) subsurface ocean temperature profiles are applied as basis for the final reconstruction. After the neural network reconstructions, the temperature profiles are converted into a two dimensional OHC map via integration:

$$H = c_p \int_{z_0}^{z_1} \rho_z \cdot T_z dz, \tag{2.1}$$

In this case H denotes the ocean heat content, c_p the specific heat capacity of sea water, ρ_z the density of the water at depth z and T_z the temperature at this depth. This OHC field is then reduced to one value by summing over all relevant gridpoints for the calculation of an OHC timeseries.

2.1.1. Data Assimilation

In general, Kalman Filter (KF) assimilations assume the uncertainties of model and observations to be Gaussian and then assimilates the observations sequentially. Both, model and observations, are described as probability density functions (PDFs) and the Kalman filter alternates between forecasting and analyzing the PDF's covariance and mean Carrassi et al., 2018. By introducing an ensemble of model trajectories instead of one single representation of the model dynamics, the KF can be adjusted and extended in what is called an ensemble Kalman filter (EnKF). This allows for the covariance matrices to be approximated by an ensemble of model states. The covariance matrix and ensemble mean are then sequentially forecast and updated.

For this work, an ensemble coupled MPI-ESM assimilation (Brune and Baehr, 2020; Brune et al., 2021) is used to train the U-net to reconstruct masked temperature profiles. An oceanic localized EnKF, implemented with the Parallel Data Assimilation Framework (Nerger and Hiller, 2013), assimilates monthly EN4 temperature and salinity profiles. Simultaneously, atmospheric nudging of ERA40/ERAInterim/ERA5 reanalyses (Dee et al., 2011; Hersbach et al., 2020; Uppala et al., 2005) takes place. Monthly mean temperatures are taken from the assimilation from January 1958 – October 2020 down to a depth of 2.000 m in 40 depth layers.

2.1.2. Observational Data

The EN4 project describes a collection of global ocean temperature and salinity profile datasets spanning from 1900 until present day. It contains observational data from Argo floats, Arctic Synoptic Basinide Oceanography (ASBO), the Global Temperature and Salinity Profile Program (GTSPP) and the World Ocean Database (WOD13) (Good et al., 2013). As these datasets partly overlap, input data is preprocessed and duplicates sorted out before the complete EN4 dataset is created. The same procedure is applied to profiles that are not necessarily identified to be the same but that are very close to each other in time and space (Good et al., 2013). Additionally, various quality control checks are performed in order to identify data profiles with faulty location or measurement data, such as missing depth steps or the location assignment to a point on land rather than in the ocean. The EN4 temperature profiles possess the same vertical and horizontal resolution as the assimilation data. All datapoints are brought together in NetCDF files and are publicly available under `https://www.metoffice.gov.uk/hadobs/` (Good et al., 2013).

2.1.3. Other OHC Estimates

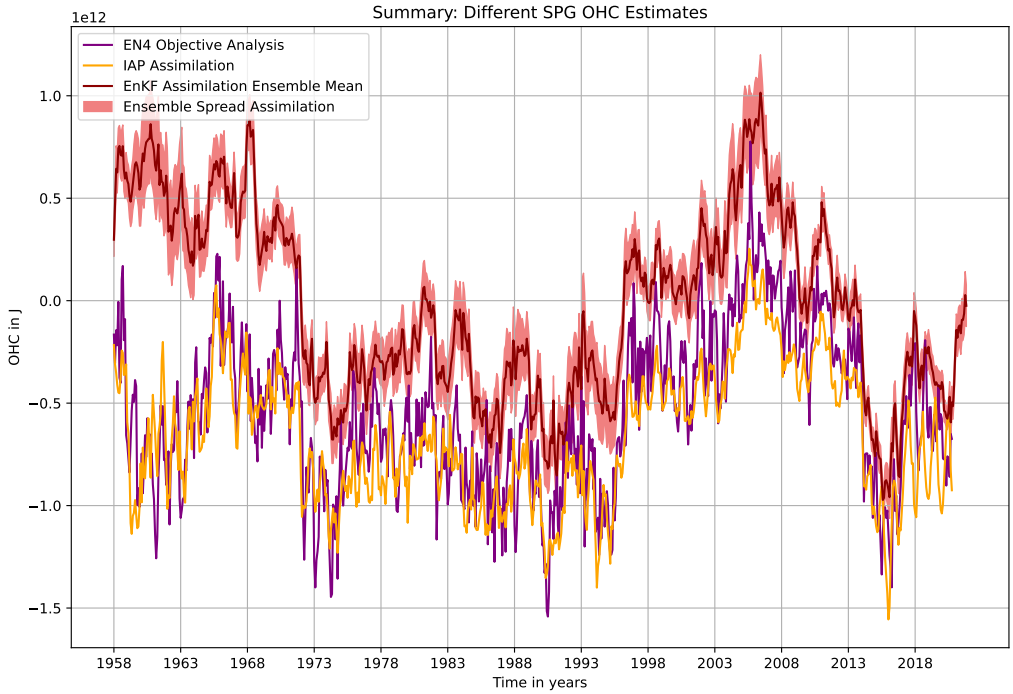

Figure 2.1.: *Comparison of EnKF assimilation ensemble mean (red) with IAP (orange) and EN4 (purple) SPG OHC estimates. The shaded area around the assimilation ensemble mean represents its ensemble spread.*

In order to contextualize the reconstruction results outside of the EnKF assimilation, two other OHC estimates will be taken as reference points. Figure 2.1 shows these in comparison to the EnKF assimilation ensemble mean and its spread. The EN4 project provides an interpolated version of the observed data points in the form of an objective analysis. This analysis is created

by combining a persistence based forecast from the previous month's analysis with the observed temperature profiles Good et al., 2013. As the persistence relaxes towards the baseline climatology when not corrected by observations, the EN4 reanalysis tends to posses a lot of uncertainty in times with low observational density. The applied baseline climatology is taken from the years 1970 – 2000. Thus, the overall cooling trend in the NA during that time (2.1) leads to a cool bias in the EN4 OHC estimate especially in the early pre-Argo era Good et al., 2013. Despite these shortcomings, the EN4 analysis will be taken as a reference point in the evaluation of the network reconstructions as it is highly consistent with the observational data, especially in the well observed Argo era. Therefore, it can help provide additional context in particular to the direct neural network reconstruction of EN4 observations.
In order to better contextualize the reconstruction results, the Institute for Atmospheric Physic's (IAP) OHC estimate is additionally provided as a point of comparison. The IAP mainly uses the same sources for its observational input as the EN4 objective analysis (Cheng and Zhu, 2016). The main difference to the EN4 reanalysis consists in the mapping of the unobserved gridpoints. In opposition to the statistical mapping of the EN4, which is only supplemented by an estimation of covariances and a relaxation term towards the baseline climatology, the IAP also incorporates model data to reconstruct missing data points. The IAP combines CMIP5 multi-model simulations with ensemble interpolation methods, which produces a data assimilation that is less biased even in regions with low observations compared to simpler interpolations Cheng and Zhu, 2016. According to Cheng and Zhu, 2016, the IAP OHC estimate provides reliable results without a significant bias from 1955 on. However, it only agrees with estimates of top-of-atmosphere radiative imbalance since 1985. Therefore, all results before 1985, but especially before 1955 possess larger uncertainties. All IAP temperature fields are publicly available and have been downloaded from `http://www.ocean.iap.ac.cn/pages/dataService/dataService.html?languageType=en&navAnchor=dataService`.

2.2. Methods

2.2.1. The Convolutional U-net

The U-net adopted in this work is based on the one used by Kadow et al., 2020 to reconstruct near-surface air temperature data. Kadow et al., 2020 adopted image inpainting technology developed by Liu et al., 2018. This original U-net focused on infilling irregular holes in a variety of images. The concept of infilling irregular holes aims at producing smooth results consistent with the holes' environment without limiting itself to rectangular holes. Liu et al., 2018 achieved that by designing a neural network, which utilizes partial convolutions. Partial convolutions consist of a masked and re-normalized convolution operation, which is then followed by an automatic mask-update step (Liu et al., 2018). In mathematical terms, this means that the partial convolution at each location is defined as

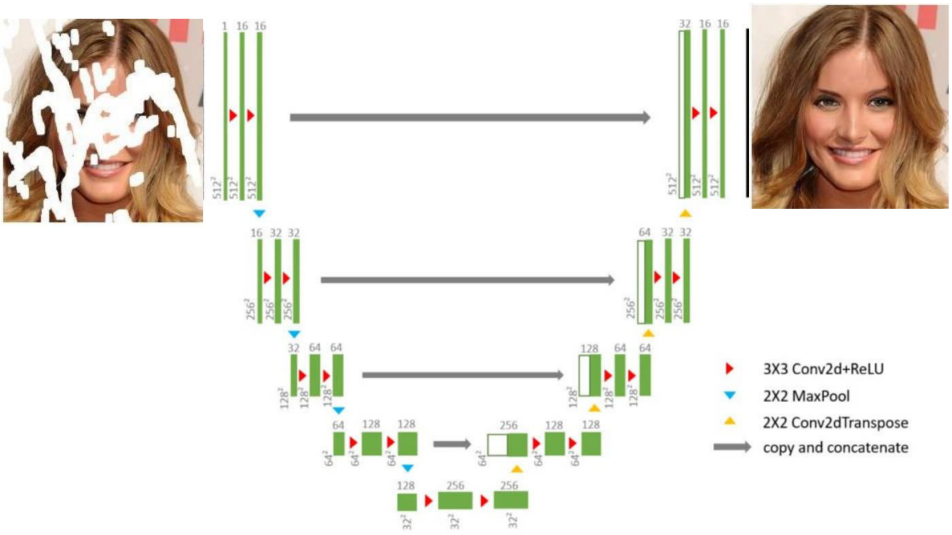

Figure 2.2.: *Depiction of the original U-net utilized by Liu et al., 2018 for the inpainting of irregular holes. In the case of this thesis, the masked images of photographs are replaced by masked assimilation data or EN4 observations.*

$$x' = \begin{cases} W^T(X \cdot M)\frac{\sum(1)}{\sum(M)} + b, & \text{if } \sum(M) > 0 \\ 0, & \text{otherwise} \end{cases}, \tag{2.2}$$

where W denotes the convolutional filter weights, b its corresponding bias, X the feature values for the current convolution, M the corresponding mask and 1 an array with the same size as M filled with ones Liu et al., 2018. This mask is then updated for each element according to Liu et al., 2018 as

$$m' = \begin{cases} 1, & \text{if} \sum(M) > 0 \\ 0, & \text{otherwise} \end{cases}. \tag{2.3}$$

With enough partial convolutions, the mask is eventually updated to incorporate only ones and the output image does not possess any empty grid points anymore. Liu et al., 2018 incorporated partial convolutions into the bigger framework of convolutional U-nets. U-nets downscale their input images with convolution and pooling operations in order to filter out the import data contained in its input in a so called encoding path. Afterwards, the downsampled data is brought back to its initial resolution through interpolation and deconvolution operations in a decoding path. This setup, where data is first condensed into a sort of bottleneck and upscaled again afterwards, is depicted in figure 2.2. The name U-net originates from its characteristical shape.

In each encoding and decoding layer of the U-net a partial convolution operation is performed. This approach yields significant upsides compared to normal convolutional layers. Convolutional U-nets rely very heavily on the initial values given to the U-net in the places of the image holes.

This often makes computationally expensive postprocessing necessary to compensate for the artifacts stemming from this initial value bias (Liu et al., 2018). Partial convolutional layers account for the initial value problem by using the described masked convolution operation followed by the automatic mask update step. This results in the reconstruction being only dependent on the input data that is not masked out and thus renders the initial hole value irrelevant (Liu et al., 2018). An additional upside of this approach is the fact that this dependence on only the valid input data renders the shape of the holes irrelevant. Both problems are also applicable to the reconstruction of missing temperature observations, as the missing data usually differs at every timestep in its shape. Thus, these partial convolutional U-nets can learn spatial patterns of temperature data and apply them to highly varying observational patterns Kadow et al., 2020.

Each layer of the U-net is activated using a Rectified Linear Unit (ReLU) activation functions. ReLU activation functions provide an easy possibility to reduce the amount of active neurons in each layer, which significantly speeds up neural networks compared to sigmoid or tanh activation functions. The U-net's training loss is calculated at each iteration by separating the overall loss function into five parts, each for a different feature of the network output. The most impactful factors in connection to the U-net's overall performance when reconstructing missing climate information are the gridpoint wise loss functions for the infilled holes and originally unmasked data points (Plésiat, 2022). The individual components are always calculated as mean absolute error (MAE) between the network output and the ground truth of the training data. The weights of the individual loss components can be adjusted and are determined manually by tryouts without any specific hyperparameter training.

2.2.2. Neural Network Setup and Parameters

The U-net conceptually described in section 2.2.1 can be setup in a number of different ways depending on the exact purpose of the training and the input data. In contrast to Kadow et al., 2020, this work focuses on three dimensional temperature profiles instead of just SSTs. Thus, instead of one network, there are several at work, with each one training to reconstruct an individual depth layer. After each layer is reconstructed, the resulting filled in temperature slices can be reassembled and postprocessed. As the aim is to calculate the upper 700 m ocean heat content, the first 20 depth layers of the assimilation training data are selected, which represent temperature fields up to a depth of 700 m.

The hyperparameters for each U-net, including network depth and loss parameters can be adjusted individually previous to each training instance. The default setup consists of four encoding and four pooling layers because until now, a further deepening of the neural network slows down calculations without providing significantly better results. Furthermore, the time period as well as the spatial extension in the form of longitudes and latitudes of the input data can be individually determined for each training process. In regard to the time period of the training data, two different arguments can be adjusted. On the one hand, the origin of the data itself can be adjusted, with default options from the Argo era, pre-Argo era or the entire assimilations time period. On the other hand, the time of origin for the observational patterns, with which the

assimilation data is masked during the training process can be changed. Thus, it is possible for the network to learn Argo-era assimilation patterns while also learning to adjust to pre-Argo era observational density.
The remaining parameters, such as learning rate, batch size and image size, are adopted from the original network set up by Kadow et al., 2020 and can be viewed with the remainder of the code under `https://github.com/slenta/Master-Arbeit`.

2.2.3. Data Preprocessing

The U-net described in section 2.2.1 takes three input images in the form of a ground truth assimilation, a binary mask and a masked out assimilation image formatted as an hdf5 file. In order feed the NetCDF4 files containing the assimilation data into the network, they have to be preprocessed to fit the given format. At first, the NA region is cut out of the global assimilation runs. This leaves an irregular 107x124 grid of the NA region from -65 – -5 °W and 20 – 69 °W. For the neural network to be able to process this data, rows of zeros are added until a quadratic 128x128 grid of the NA is reached. Then, the assimilation's temperature profiles are converted to anomalies using the assimilation's average monthly climatology. For this calculation only the time period from 2004 – 2020 is considered, as this work assumes that the assimilation possesses significantly less uncertainty during the well observed Argo era. Then, EN4 observations from a selected input time are converted to binary masks and multiplied with the data assimilation model data. All three, the masks, the masked data as well as the data assimilations are then stored in an hdf5 file. From there, they can be loaded, split up into batches and fed into the network using a class with the *PyTorch* dataset module.
After the training process, when reconstructing a dataset with monthly observations for a certain timespan, the U-net returns outputs with fully reconstructed monthly temperature profiles.

2.2.4. Analysis Approach

After the network has been fully trained, visible at the progression of its training curve, it can be given masked temperature profiles for reconstruction. The network will not be trained with all 16 ensemble members. Instead, 4 ensemble members will be held back for later evaluation. This process helps to ensure that the network does not commit to overtraining on its training data. At first, the network will reconstruct the Argo era masked assimilations from one of the held back ensemble members. Each month of assimilation data will be masked with the corresponding observational pattern from that time. The resulting reconstructed OHC can then be summed up over relevant areas and visualized in the form of a timeseries. While the network is trained with the masked assimilations from the entire NA, the data for all other analysis steps is limited to just the SPG. Thus, the neural network is given the opportunity to also learn the SPG surrounding spatial patterns, while the remaining analysis is focused on a region that is of higher interest to current research. Therefore, the results can also be better compared to other OHC estimates. Given the high observational density during the Argo era, this reconstruction can be considered the first step for the neural network towards the reproduction of the assimilation OHC estimate. This approach can then be extended to the pre-Argo era, if the neural network is able to reproduce the assimilation ensemble within its uncertainty range during the Argo era.

Analyzing the reconstructed OHC timeseries as well as maps from individual timesteps, the machine learning reconstructions can also be used to discuss inconsistencies and uncertainties within the assimilation. Differences between both OHC estimates can then be taken as indicator for uncertainties that are not mirrored within the assimilation's ensemble spread. Finally, I will examine, whether the neural network is able to directly reconstruct observations in the pre-Argo and Argo era as well as more recent observations it has not trained on within the assimilation's uncertainty range. The results will then be evaluated in regard to the neural network's potential to reproduce or replace EnKF assimilations. Additionally, its implications on a new uncertainty estimate on the EnKF assimilations OHC outside of its ensemble spread will be considered.

3. Results

3.1. Present: Argo-Era Reconstructions

At first, the U-net is trained with assimilations as well as masks only from the Argo-Era. Thus, it is set up as well as possible to reproduce Argo-Era assimilations. The training process is then conducted until the loss parameters do not show any significant decrease anymore. Manual selection has shown that the most important loss parameters for the reconstruction of masked temperature profiles are those monitoring the direct misfit of each gridpoint and the total variation loss between neighboring gridpoints. Thus, their multiplicators have been upgraded in this version of the convolutional U-net and their development will be shown to indicate the state of training of the neural network.

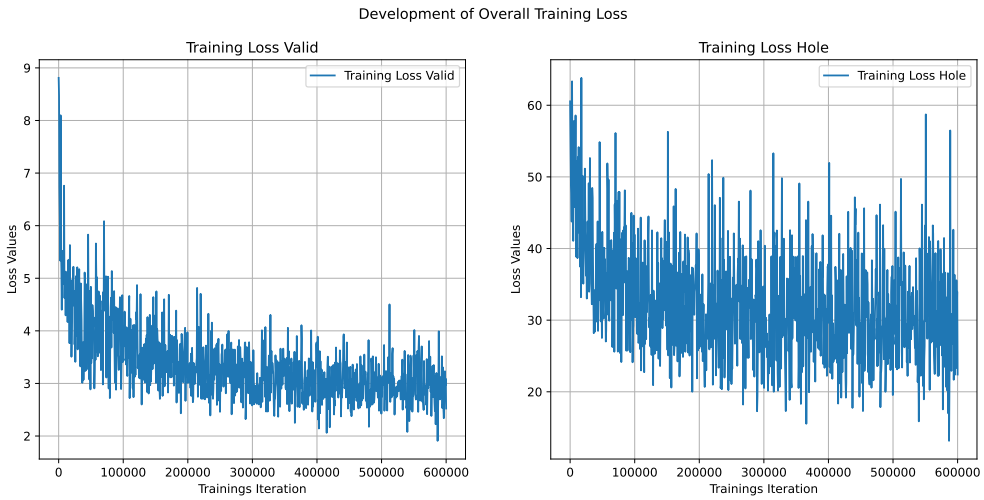

Figure 3.1.: *Mean Hole and Valid training loss for all depth layers over 600.000 training iterations of the neural network.*

After 600.000 iterations, both the hole and valid loss depicted in figure 3.1 have reached a plateau with the hole loss showing a higher variability. The loss curves for all individual depth layers are attached in section A. An analysis of both loss functions shows that the network state with the lowest combined loss occurs at approximately 585.000 iterations. The state of the neural network at that loss minimum is therefore utilized for further analysis.

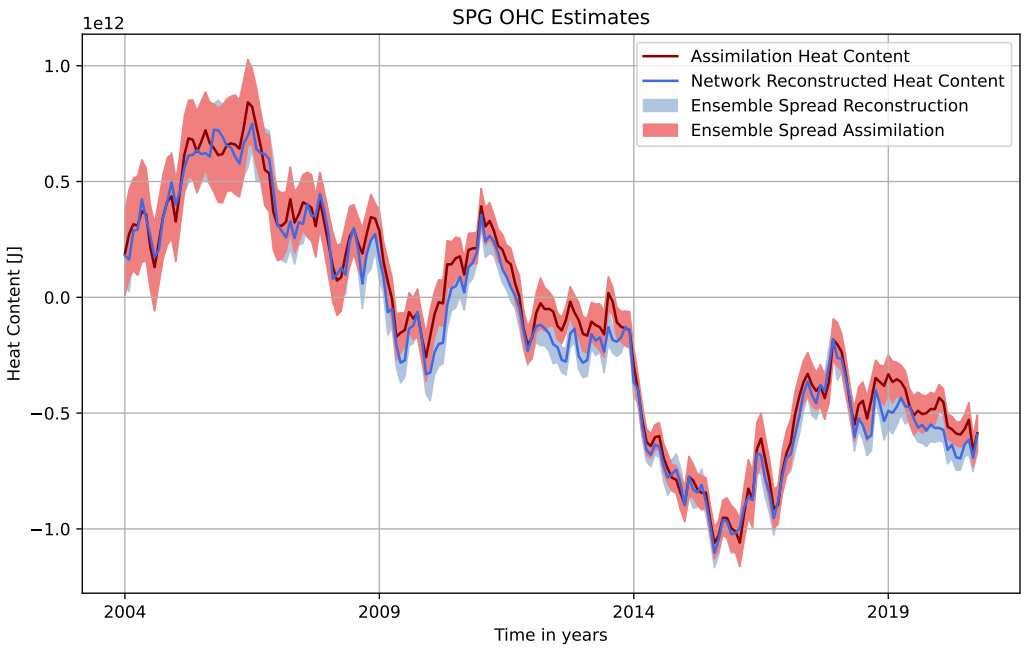

Figure 3.2.: *Estimates of upper 700 m SPG OHC of one EnKF assimilation's ensemble member (red) and the neural network reconstruction (blue) during the Argo-Era. The shaded areas represent the uncertainty caused by the corresponding ensemble spread. In the case of the neural network, this ensemble is produced by reconstructing each assimilation ensemble member individually.*

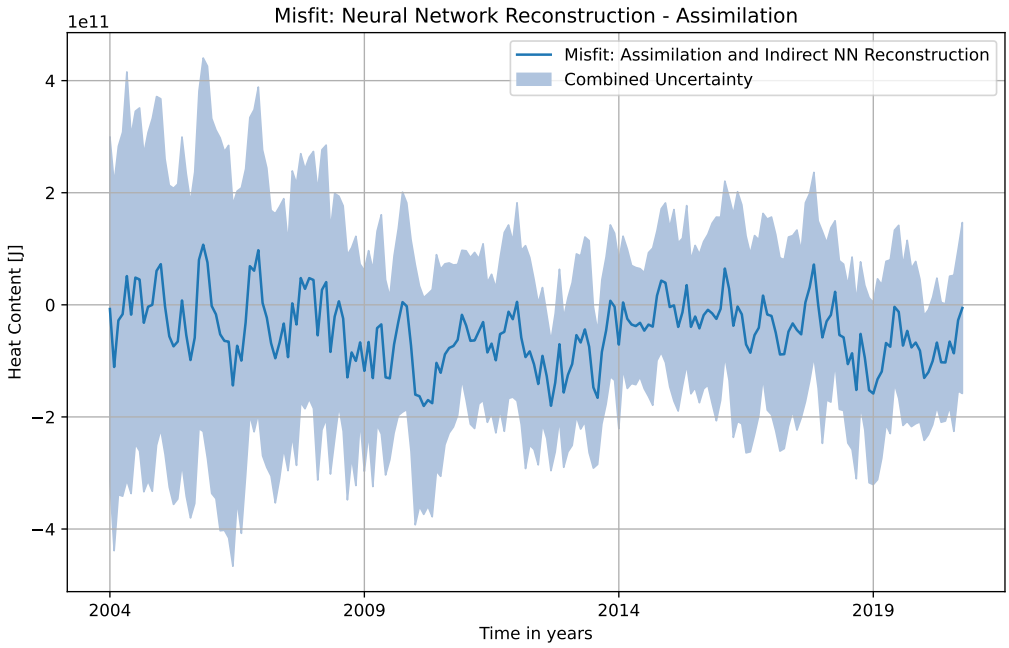

Figure 3.3.: *Misfit between EnKF assimilation NA upper 700m OHC and the neural network's reconstruction.*

Within the Argo-Era, the neural network is able to reconstruct assimilations masked with observational patterns within the assimilation's uncertainty range for monthly values (figure 3.2). Not only do both uncertainty ranges overlap, with a correlation of 0.99 the fit is clearly better than just inside of one standard deviation. The neural network's reconstruction even manages to capture monthly trends in the assimilation very well. Analyzing this timeseries further, the misfit between both estimates depicted in figure 3.3 shows that between 2009 and 2014 the reconstruction possesses a slight cool bias compared to the assimilation. But, as the remaining timeseries does not show any clear bias but rather just errors distributed around zero, it can be assumed to be insignificant and not a sign of a larger uncertainty within the network reconstructions. Figure 3.3 also shows that even during the Argo-Era, the uncertainty of the OHC estimates significantly declines in the first few years before the Argo floats reached global coverage.

Figure 3.4 further underlines the great agreement between both OHC estimates. It shows the NA OHC as calculated by the assimilation and neural network in January 2008. The corresponding input masked assimilation is shown in section A. Not only does the network's overall OHC agree with the EnKF assimilation, but also all major spatial OHC patterns are represented well by the neural network. Even on this monthly scale, the neural network is able to capture small scale features, for example at the North African coast, in the North-Western NA or even most of the eddy like structures of the North Atlantic current in the south of the SPG.

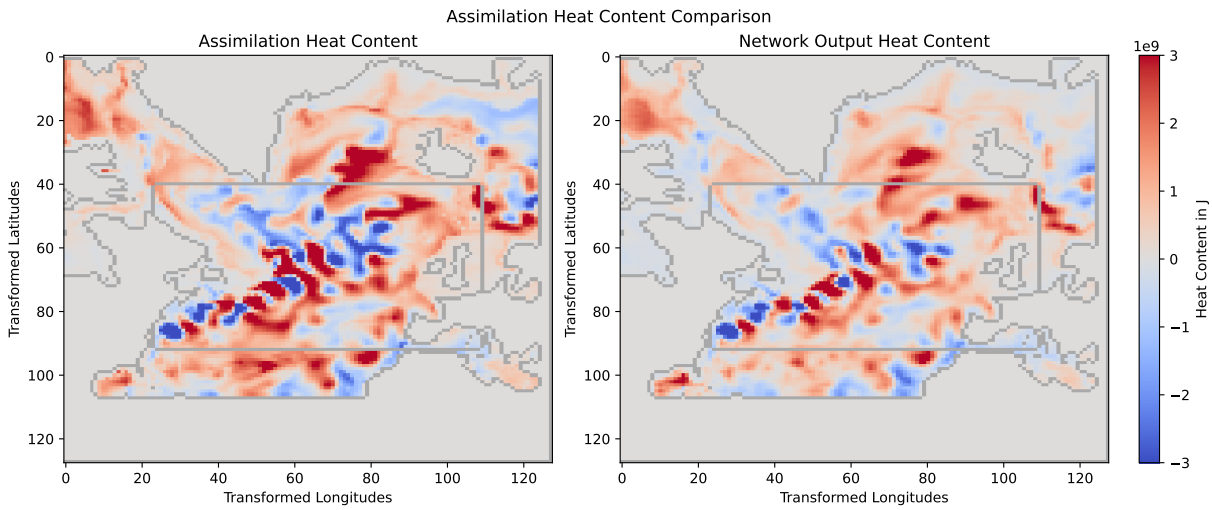

Figure 3.4.: *Map of EnKF assimilation (left) and neural network reconstruction of NA upper 700m anomaly OHC estimate in January 2008. The SPG region that is used for the construction of the timeseries is marked by the grey rectangle. The unusual projection of the North Atlantic is oriented at the assimilation's irregular model grid, which enlarges the SPG region.*

To further analyse the neural network's capability to capture the assimilation's spatial patterns, the pattern correlation between both OHC estimates can be taken. The pattern correlation is calculated by flattening both OHC images to a one-dimensional array and taking the anomaly correlation coefficient of both arrays. Thus, the similarities between spatial patterns for both OHC images can be estimated. Figure 3.5 depicts this pattern correlation during the Argo-Era. During the first years of the Argo-Era both OHC estimates show a slightly lower pattern cor-

relation while with rising observations during the Argo-Era the pattern correlation reaches a constant value of over 0.8 with an overall mean pattern correlation of 0.86. The first 5 years also show higher month-to-month variability in pattern correlation. This can also be attributed to the higher inconsistencies in data coverage during the first years of the Argo project.

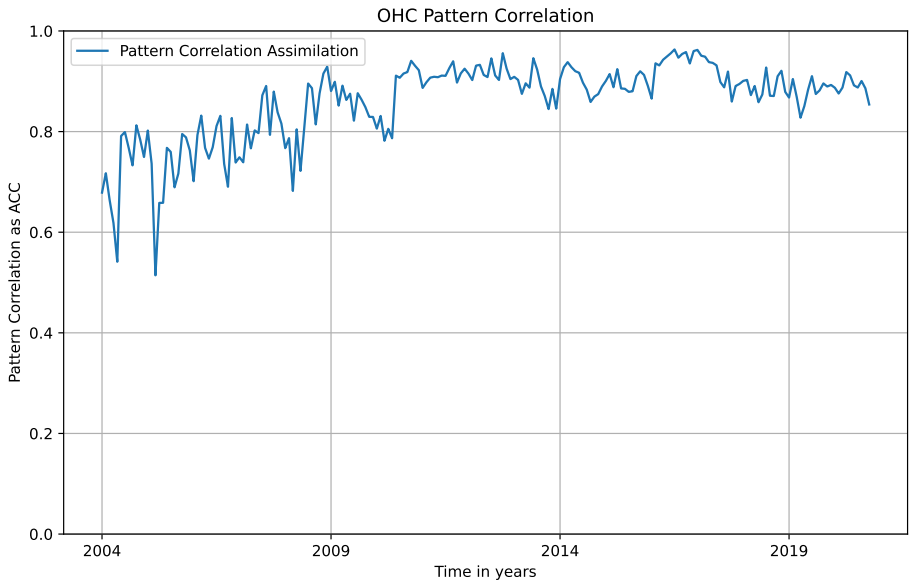

Figure 3.5.: *Pattern correlation for monthly estimates of anomaly NA OHC of one EnKF assimilation ensemble member and neural network reconstructions of masked assimilations.*

3.2. Past: Extension to pre-Argo-Era

In a next analysis step, the time window of reconstruction is extended to the pre-Argo-Era. In order to prepare the neural network for the reconstruction of the observational patterns of the pre-Argo-Era, the training process has to include pre-Argo masks. Therefore, the training data, which still only stems from the Argo-Era, is masked with observational patterns from the entire period from 1958–2020. Otherwise, the neural network's parameters are kept constant.
Even after 1.000.000 training iterations, the network's loss functions over all depth layers are still decreasing significantly (figure 3.6). The loss developments of all individual depth layers show very similar behavior and can be further examined in chapter A. Usually, these training curves would indicate that the training should be resumed until a stable plateau is reached. However, the overall validation error, which depicts the mean mismatch between the OHC timeseries of the held back evaluation assimilation member and the neural network reconstruction, suggests that no further improvement in the reconstruction accuracy occurs (figure 3.7). Instead, utilizing a network state around iteration 550.000 might yield the best reconstruction results. This stopping point also agrees with the loss minimum in the hole loss function. Thus, this network state after 550.000 iterations is utilized for the following analysis.

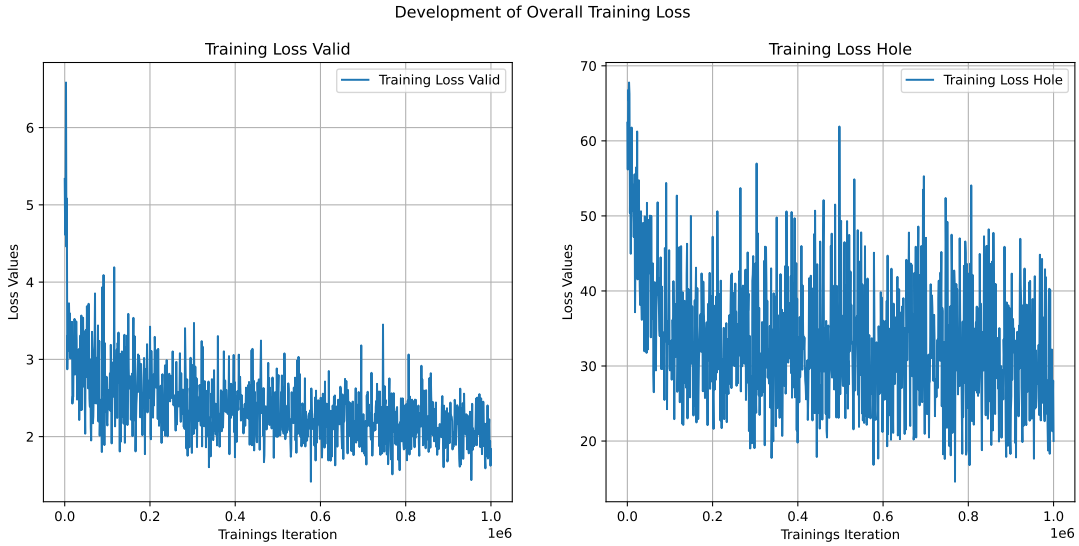

Figure 3.6.: *Hole and valid training loss for the first 1.000.000 training iterations of the neural network.*

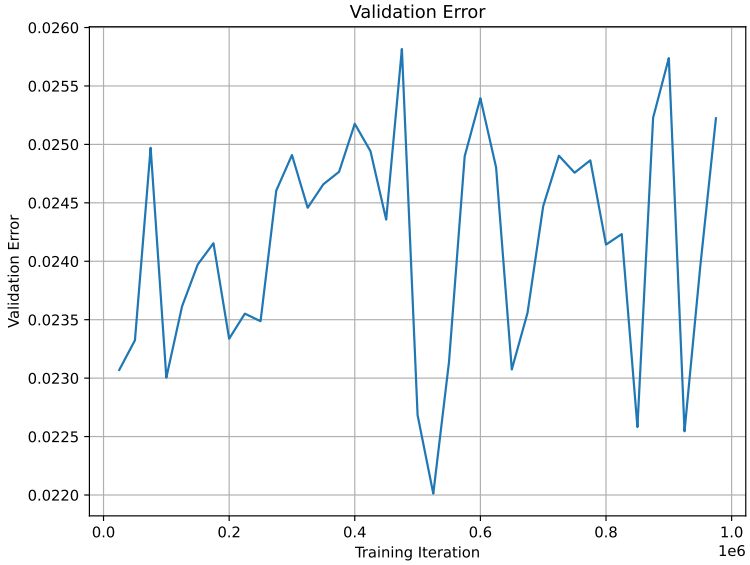

Figure 3.7.: *Mean timeseries mismatch between the evaluation assimilation member and the neural network reconstructions for the first 1.000.000 training iterations of the neural network.*

The results of the following reconstruction, which can be viewed in figure 3.8, show higher disagreements between both OHC estimates in the pre-Argo-Era compared to the previous Argo-Era reconstruction. Nonetheless, with an anomaly correlation of 0.93, they still agree very well with each other. In comparison to the first part of the analysis, this plot shows the annual running means of both OHC timeseries. During the pre-Argo-Era, the monthly fluctuations in observational coverage lead to higher variability in both assimilation and neural network reconstruction. Taking the running mean of both timeseries clearifies the analysis, because outlying months, which are results of very low amounts of observations in unusual regions, distort the

image less.
Supported by the depiction of the misfit between both timeseries in figure 3.9, it becomes obvious that the difference between both timeseries consists of more than just increased variability due to higher uncertainty caused by less observations. Instead, some trends become obvious in this annual timeseries. Firstly, the assimilation shows much higher interannual variability and a clear peak in OHC that the reconstruction does not capture during the first ten years of the timeseries. Secondly, the neural network shows a consistent warm bias during the following 20 years. The two main options as causes for these differences are on the one hand an overfitting of the neural network to the training period. The possibility that the neural network does not recognize assimilation patterns from the beginning of the pre-Argo-Era, if they do not resemble those of the training period, has to be considered. On the other hand, it is also possible that the differences are caused just by the shift in observational density between the Argo and pre-Argo-Era. Then, these differences could be utilized as indicators of uncertainties in the assimilation beyond its ensemble spread.

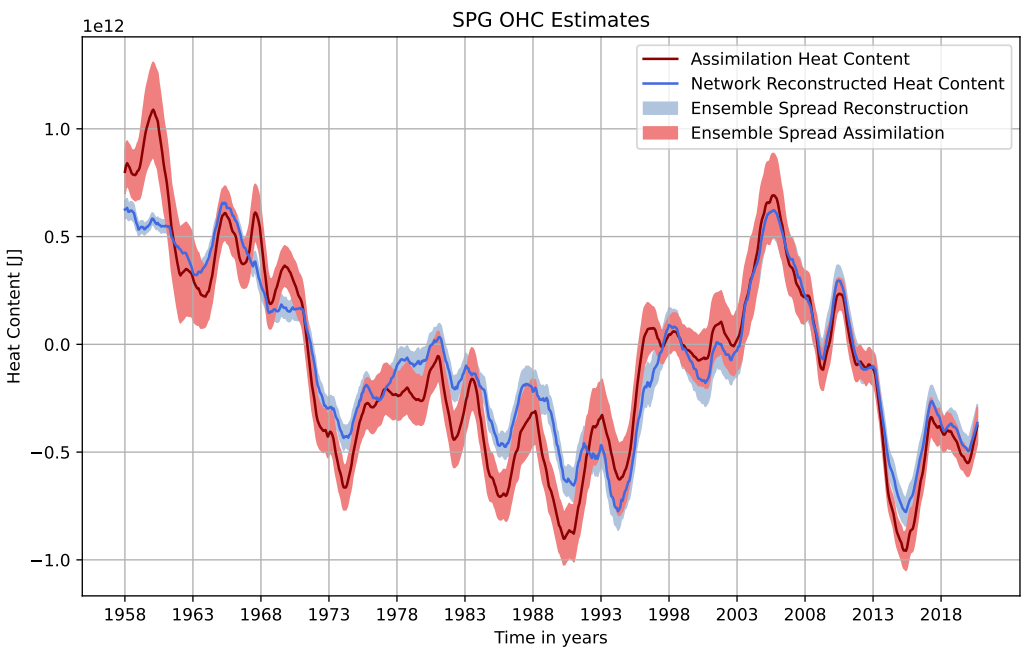

Figure 3.8.: *Estimates of annual running means of upper 700 m North Atlantic OHC of one EnKF assimilation's ensemble member (red) and its neural network reconstruction (blue) during the pre-Argo and Argo-Era. The shaded areas represent the uncertainty caused by the corresponding ensemble spread.*

Figure 3.10 supports this second hypothesis. It shows reconstructions over the same period, with the difference that the observational patterns of the Argo-Era are also used during the pre-Argo-Era to mask the assimilations. Therefore, this reconstruction gives an estimate on how much of the reconstruction uncertainty comes from the amount of observations. However, it also prevents the neural network from implementing assimilation patterns from the Argo-Era in the reconstruction of the pre-Argo assimilations. By using the Argo-Era observational patterns,

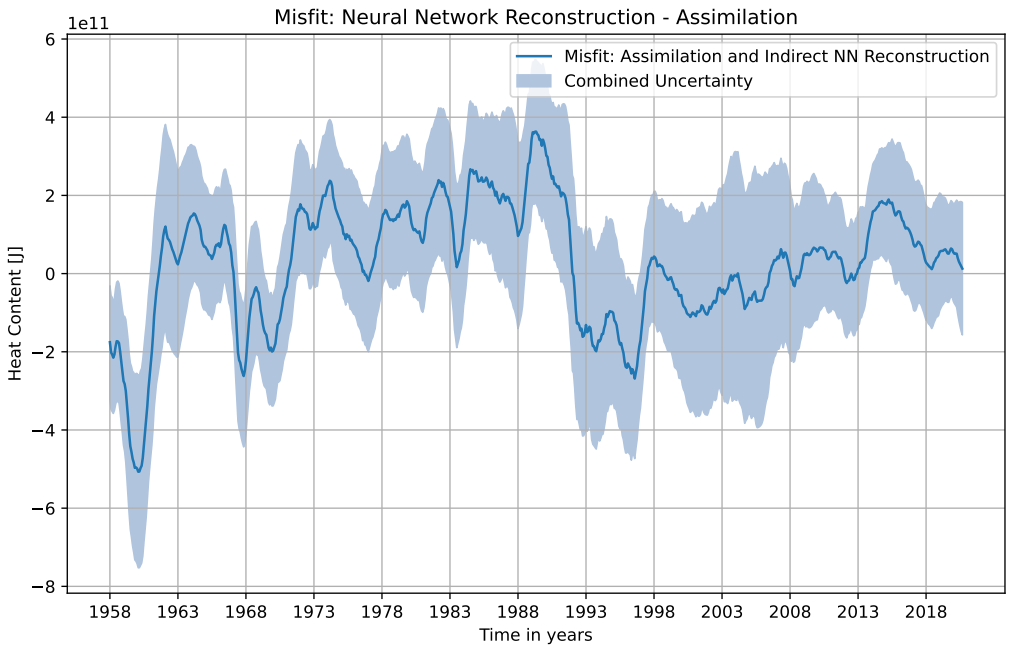

Figure 3.9.: *Misfit between EnKF assimilation NA upper 700m OHC and the neural network's reconstruction during pre-Argo and Argo-Era.*

the high observational density pushes the neural network very close to the assimilation without leaving it its own room for interpretation. One of the main assumption of this thesis is that the assimilation is more accurate over the training period. Therefore, the difference between reconstruction and assimilation cannot be interpreted as a fault of the neural network. Instead, it rather shows the uncertainty the assimilation incorporates, if the underlying model cannot be supported and nudged by sufficient observations.

Closer examination of the reconstructions and assimilation during the first decade of the time-series can help explain the observed uncertainty in that time period. In figure 3.8 the assimilation shows much stronger interannual variations in SPG OHC at the start of the 1960s than the neural network reconstructions. Koul (pers. comm.) has already examined similar EnKF assimilations during that period and found that the assimilation seems to possess much higher uncertainty during that time than its ensemble spread suggests. The biggest reason for this uncertainty is the dependence of the assimilation on observational input to correct for biases in the model, for example due to its positioning of the North Atlantic current. But also other uncertainties in the model get highlighted when there are no observations available to assist the assimilation in its estimate. The differences between both timeseries support that assumption. They show that the assimilation's behavior during this period is not consistent to the spatial patterns and geophysical processes during the neural network's training period. This assumption is underlined by the pattern correlation corresponding to figure 3.8, which is depicted in figure 3.11. It shows a much lower pattern correlation during the pre-Argo-Era with a mean of 0.62 before starting to increase at the start of the new century to over 0.8 in the Argo-Era resulting in a mean pattern correlation over both eras of 0.70.

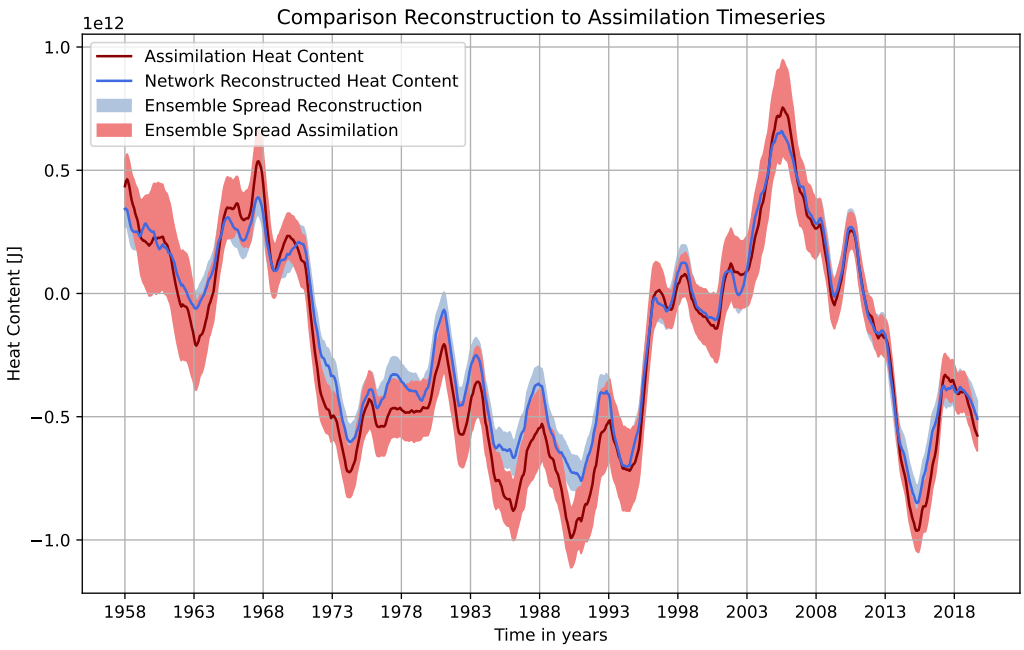

Figure 3.10.: *Estimates of monthly upper 700 m North Atlantic OHC of one EnKF assimilation's ensemble member (red) and its neural network reconstruction (blue) during the pre-Argo and Argo-Era. For the reconstruction, the assimilation member has only been masked with Argo-Era observational patterns. The shaded areas represent the uncertainty caused by the corresponding ensemble spread.*

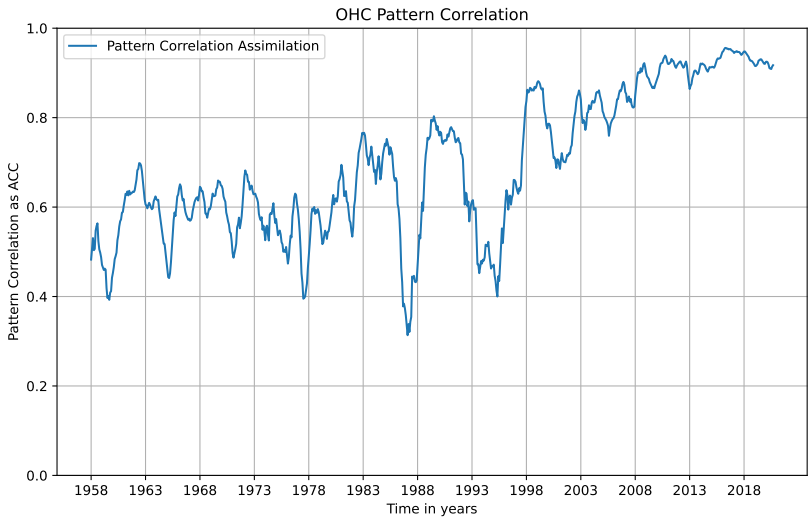

Figure 3.11.: *Pattern correlation between EnKF assimilation and the neural network indirect reconstructions during the pre-Argo and Argo-Era.*

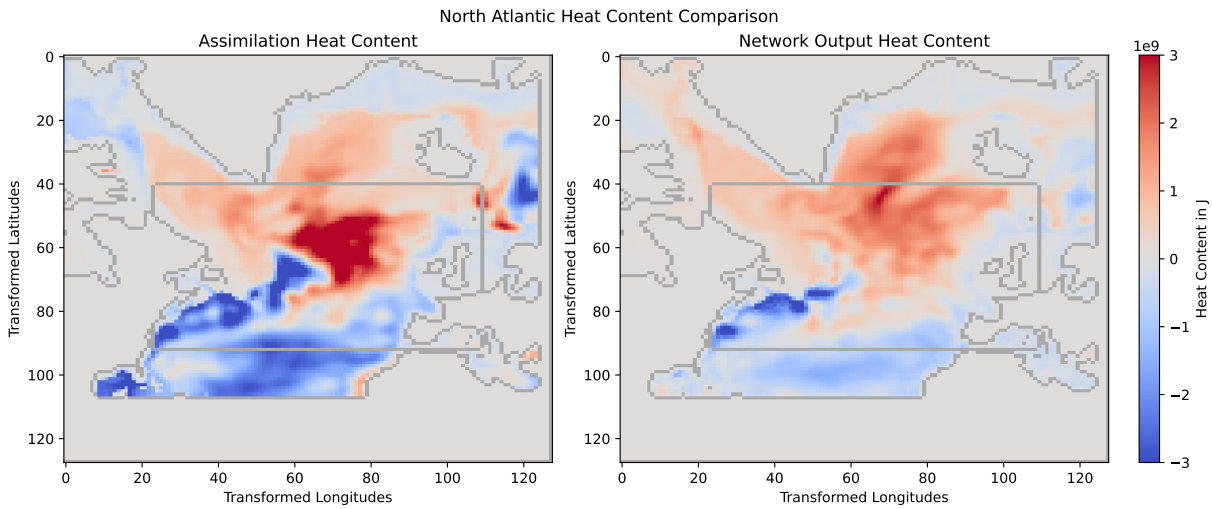

Figure 3.12.: *Map of EnKF assimilation (left) and neural network reconstruction of NA upper 700m anomaly OHC in the first ten assimilation years 1958 – 1968.*

Figure 3.12 visualizes the OHC of both the assimilation and reconstruction during that time period in the form of a 10 year mean. Both versions differ most notably in the central NA, where the assimilation shows strong negative anomalies, while the neural network reconstruction depicts close to no anomalies in that region. Similar differences have already been described in other publications and can be related to what is known as the North Atlantic Current's *North-West corner*. Many climate models, such as the MPI-ESM underlying this EnKF assimilation, are known to represent the gulf stream in the central NA further south than its observed flow. Therefore, the assimilation shows strong negative anomalies in the region of this North-West corner whenever there are no observations available to nudge the underlying model towards a correct representation of the gulf stream's flow. As the neural network is only trained in a period with high observational density in the entire NA, it only learns a correct representation of the gulf streams flow. Correspondingly, the reconstructions do not show the same negative anomalies in this region, when there are no observations available to suggest anything different.

Thus, it can also be assumed that parts of the differences between assimilation and neural network reconstruction during the first years can be attributed to model uncertainties similar to the false representation of the North-West corner. From the 1970s onward at least sporadic observations are available in the entire North Atlantic causing assimilation outputs more consistent with the neural network reconstruction. Figure 3.14 supports this analysis, as it shows that the gridwise correlation between reconstruction and assimilation is especially low in the region of the North-West corner during the first 10 years, while it depicts higher correlation in the rest of the pre-Argo-Era with the steady increase in the number of observations. The mismatch between both estimates suggests that during a time with such a low amount of observations the assimilation can be subject to model uncertainties and misrepresentations also outside of the North-West corner region. This conclusion would then mean that a neural network trained with assimilation data during a time with high observational density can be able to detect model uncertainties in specific regions. Additionally, it can provide an estimate of the overall uncertainty not contained within the assimilation's ensemble spread.

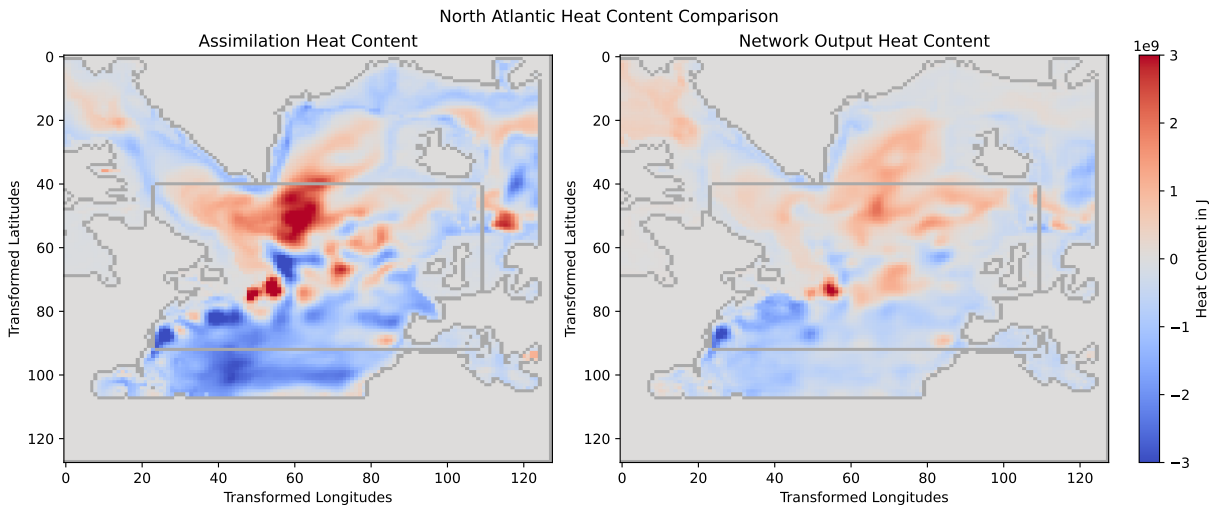

Figure 3.13.: *Map of EnKF assimilation (left) and neural network reconstruction (right) of NA upper 700m anomaly OHC in the years 1970–1975.*

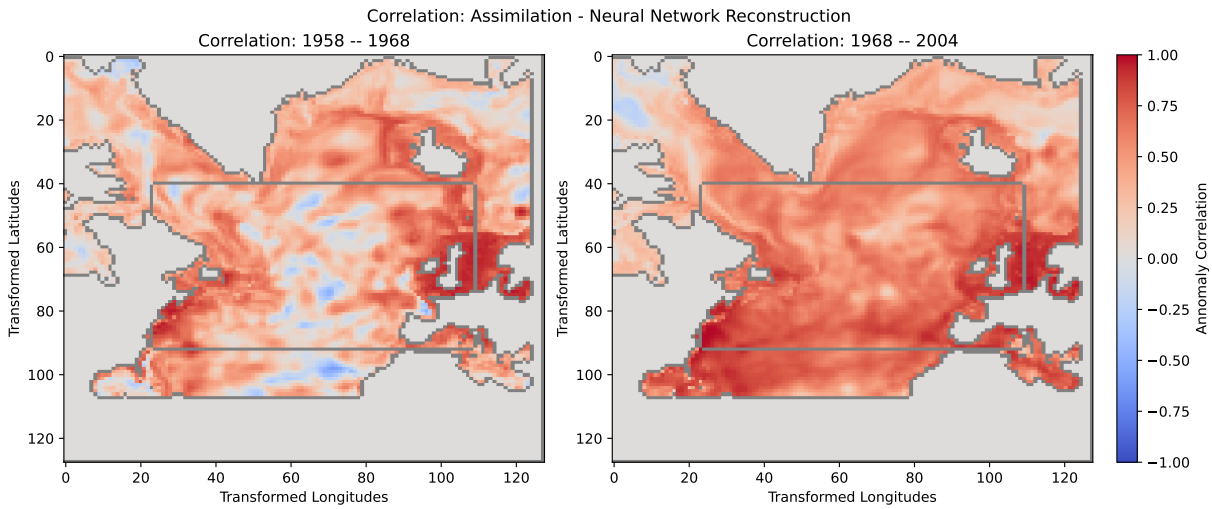

Figure 3.14.: *Grid-wise anomaly correlation between the EnKF assimilation and the neural network reconstruction of NA OHC in the years 1958 – 1968 (left) and 1968 – 2004 (right).*

During the remainder of the 20th century reconstruction and assimilation mostly agree within their respective uncertainty ranges. Nonetheless, the reconstruction shows a positive bias until the beginning of the 1990s. This positive bias could still be a remainder of similar model uncertainties related to the observational density. Afterwards, the differences vanish slowly but steadily until the beginning of the Argo-Era. From thereon out, both timeseries are again in good agreement within one standard deviation of their ensemble spreads.

3.3. Direct Observations Reconstruction

The reconstruction of masked assimilation aims mainly at assessing the neural network's ability to reproduce EnKF assimilations and identifying uncertainties in the assimilation stemming

from the underlying model. Another uncertainty that cannot be examined with this approach is the assimilation's integration of observations. Additionally, it is not possible to estimate the neural network's potential to replace data assimilation, when the neural network is never given actual observations as input. Thus, the neural network is now given the EN4 observations from 1958 until 2021 as input in a next evaluation step. To clarify the terminology, the reconstruction of observations by the neural network will from now on be called *direct reconstructions*, while the reconstruction of masked assimilations will be referred to as *indirect reconstructions*. As the direct reconstructions cannot reconstruct the assimilation ensemble to calculate its own uncertainty, a 16 member ensemble of direct reconstructions is created by reconstructing the observational input at different network states. Around the original stopping point of 550.000 iterations, the network state is extracted at 15 other iterations, each 500 training iterations apart. Thus, the resulting ensemble spread is not the same as previously with the indirect reconstructions but it does give a rough estimate of the uncertainty involved in the neural network reconstructions. A further discussion of these uncertainties is given in section 3.5.

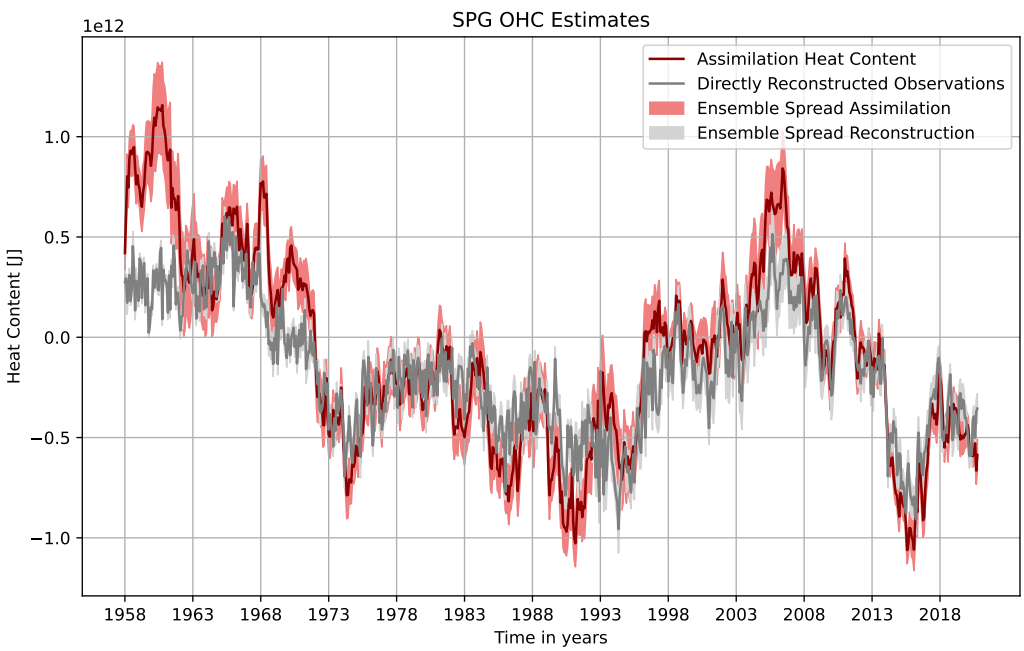

Figure 3.15.: *Estimates of monthly upper 700 m SPG OHC of one EnKF assimilation's ensemble member (red) and direct neural network reconstruction (green) during the pre-Argo and Argo-Era. The shaded areas represent the uncertainty caused by the corresponding ensemble spread.*

The direct reconstruction OHC estimate shows higher disagreements to the assimilations (3.15), but still shows high correlation with an anomaly correlation of 0.89. Two time periods stand out in particular in this figure. At first, the direct reconstruction shows clear disagreement with the assimilation during the first 10 years from 1958 – 1968. This behavior can again be related to the uncertainty of the assimilation with very few observational input already discussed in 3.2.

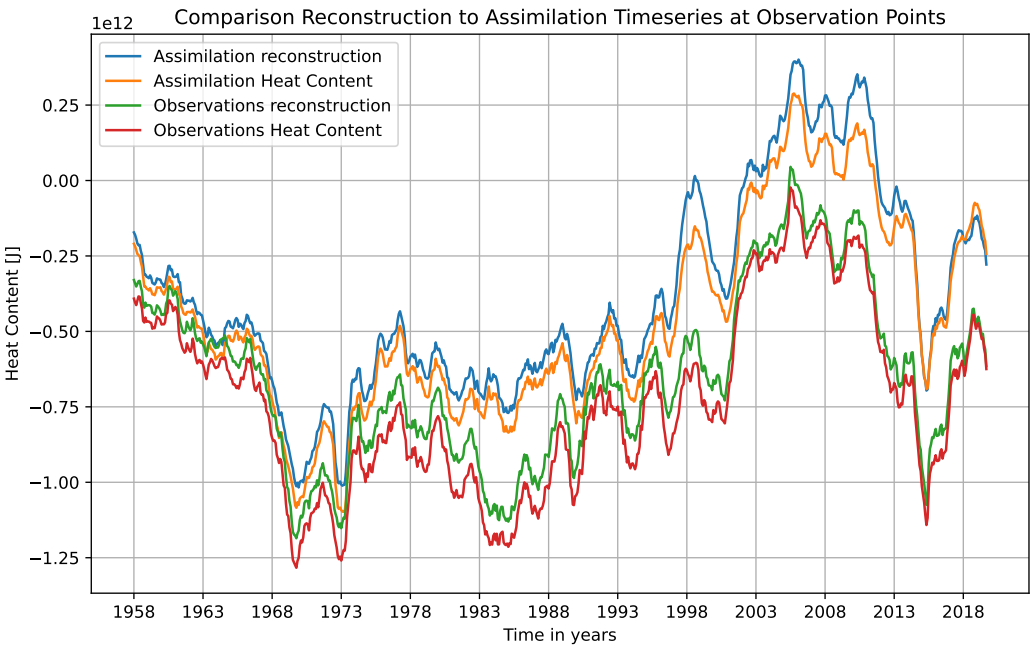

Figure 3.16.: *Estimates of upper 700m SPG OHC at the grid points of observations. Displayed are the EnKF assimilations (blue), the observations (green), the direct reconstructions of observations (red) and the indirect reconstructions of masked assimilations (orange).*

Additionally, especially during the peak around the year 2007, the direct reconstruction shows a higher deviation from the assimilation timeseries compared to the indirect reconstruction. In general, the direct reconstructions show a cool bias compared to the assimilation during most of the Argo-Era. To examine the reason for this bias and to put the direct reconstruction into the context of the EN4 observations, it is useful to compare the assimilation and the observations at the points of observations.

The OHC timeseries at the points of observations depicted in figure 3.16 shows that the observations possess a significant cool bias compared to the assimilation. The neural network reproduces its corresponding input very closely and thus it is not surprising that the bias from its input is then transferred to its overall estimation. However, the bias of the neural networks input has a smaller effect on its overall output during the pre-Argo-Era due to he smaller percentage of input points compared to the output. Furthermore, the bias between assimilation and observations is less distinct during the pre-Argo-Era. This bias is not unusual. Most assimilations show small biases towards the observations, which are caused by the model dynamics in the background. As the assimilation regards model and observations as uncertain it does not just integrate the exact observations value but finds a compromise between both factors (Brune, pers. comm.). In order to better interpret this bias it has to be put into the context of other OHC estimates.

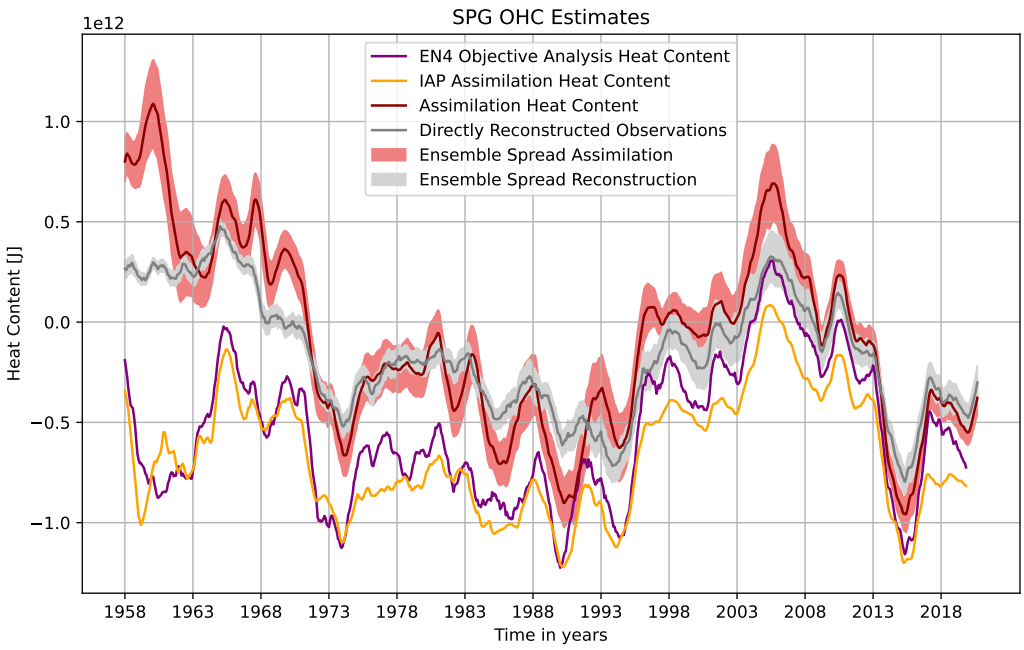

Figure 3.17.: *Estimates of annual running means of upper 700 m SPG OHC of one EnKF assimilation's ensemble member (red) and direct neural network reconstruction (grey) during the pre-Argo and Argo-Era. The shaded areas represent the uncertainty caused by the corresponding ensemble spread. In addition, the estimates of the IAP (orange) and EN4 (purple) are provided.*

The IAP and EN4 OHC estimates shown in figure 3.17 agree notably well with each other while showing a significant cool bias compared to the neural network reconstruction and the EnKF assimilation in particular. During the Argo-era, the EN4 OHC estimate even shows a higher correlation to the direct reconstruction (0.98) than the assimilation (0.96). The difference during the pre-Argo-Era however is especially large in the first 10 years, where both timeseries show opposite anomalies. This difference indicates another uncertainty in the assimilation. While the low observational density can explain some of the differences, it is not enough to explain the entire discrepancy between the assimilation and the other OHC estimates. Koul (pers. comm.) has examined the EnKF assimilation closer and has concluded that the assimilation does not deliver reliable results directly after its start. Instead, Koul (pers. comm.) has shown that the assimilation requires time to adjust to the observations after its initial start. Therefore, it has been shown to possess high uncertainties until the start of the 1970s. This adjustment period fits well to the differences to the IAP and EN4 estimates and relaxes towards a simple positive bias with the start of the 1970s. The neural network reconstruction also shows much higher disagreement to the assimilation during that period, while it also clearly does not follow the other estimates. The still existing agreement between the neural network reconstruction and the assimilation indicates that the patterns responsible for this adjustment period are also somehow existent during the Argo-Era. If they were completely independent from the remaining assimilation, a neural network trained during the Argo-Era would then show higher differences during this unrelated adjustment period.

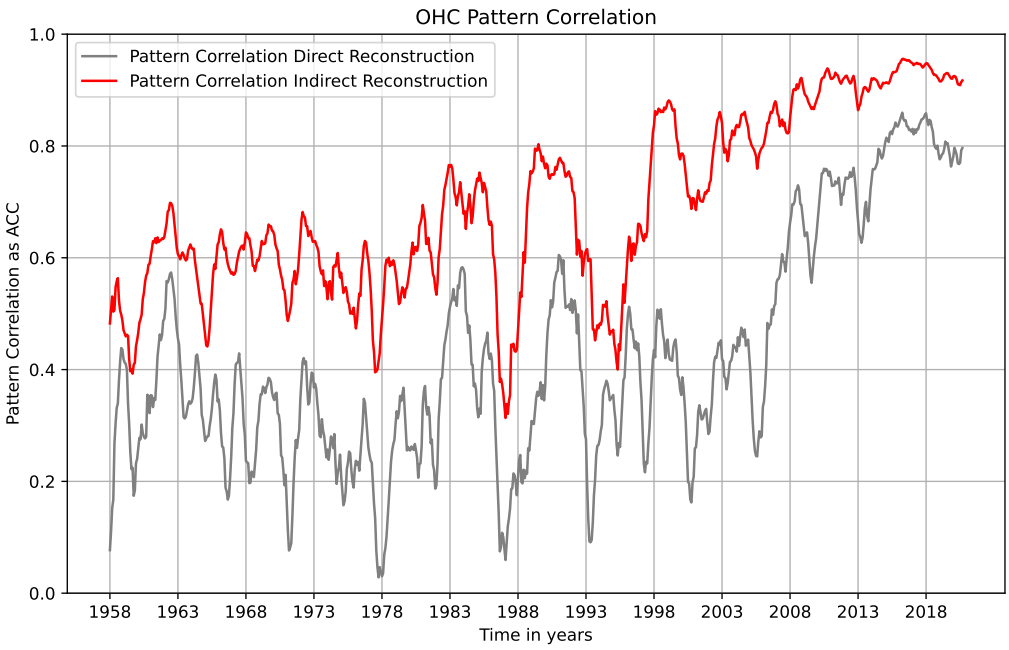

Figure 3.18.: *Pattern correlation of the annual running means of the EnKF assimilation and the direct neural network reconstruction (grey) and indirect neural network reconstruction (red).*

During the remaining pre-Argo-Era the reconstruction then follows the assimilation quite closely, showing the same warm bias in comparison to the IAP and EN4 estimates. From the middle of the 1990s until the 2010s the neural network estimates a middle way between the EN4 and the EnKF estimates. This development of the reconstruction during the Argo-Era can be read as an indicator that the warm bias in the assimilation is in fact a result of a bias within the assimilation at the points of observations. If instead it was due to a different assessment of the physical processes, the neural network trained during that period would probably agree closer with the assimilation than it does with the EN4 analysis.
When visualizing the pattern correlation between observations reconstruction and the EnKF assimilations, it becomes obvious that there are more differences between both OHC estimates than a pure bias. Figure 3.18 depicts much lower correlations in the direct compared to the indirect reconstructions with a mean pattern correlation of 0.44. The comparison to the pattern correlation of the indirect reconstruction also shows that both show very similar behavior. Therefore, it can be assumed that during the times with low pattern correlation, the observations cause the assimilation to form spatial patterns that the neural network has not learned. The difference between observations and assimilation then only induces a bias into the pattern correlation. This bias also proves that the neural network integrates the patterns from the observations instead of just translating the value bias and then forcing the observations into the same spatial patterns as the assimilation.

Figure 3.19 supports this hypothesis. It depicts the neural network's direct reconstruction of observations in 2018. However, the observations show clear regional differences compared to the assimilation. Even though the overall OHC estimates of reconstruction and assimilation agree

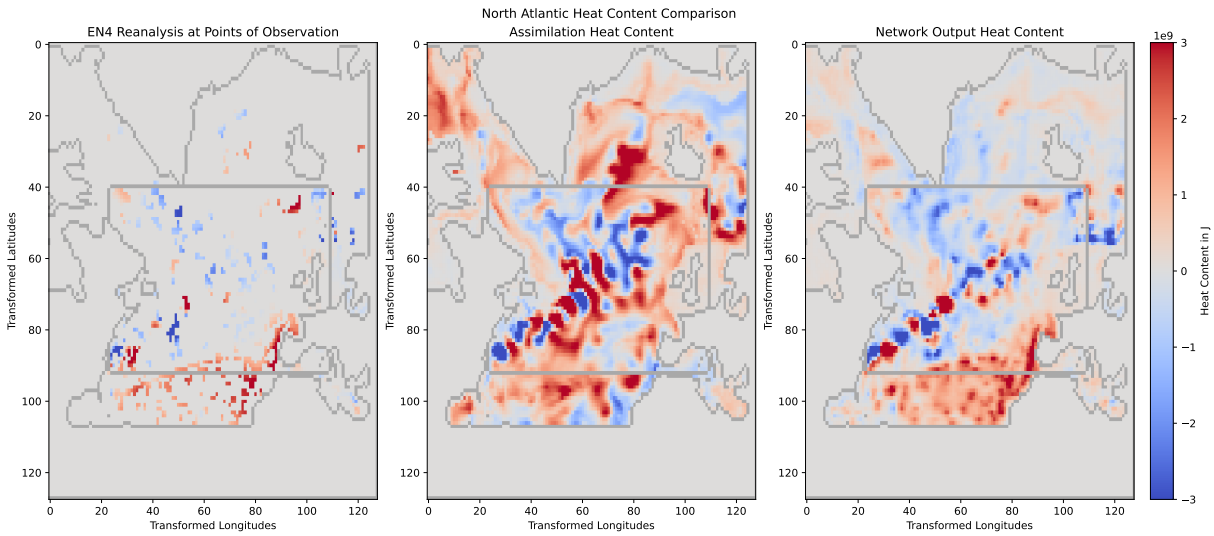

Figure 3.19.: *Map of the EN4 observations (left), the EnKF assimilation (middle) and neural network direct reconstruction estimates of NA upper 700m anomaly OHC estimate in January 2008.*

very closely at that time, the regional patterns of the observations are incorporated more by the neural network. But, the image of the direct OHC reconstruction also shows that the neural network produces less pronounced spatial patterns on smaller scales. Beyond larger structures, such as the SPG cooling and the warming of the STG it shows very few smooth small scale patterns. So even though the neural network seems to replicate large scale observational patterns, it has difficulties in the resolution of smaller scale structures. This difficulty to produce smooth regional patterns is most likely also the reason for the low pattern correlation of the direct reconstruction during most of the pre-Argo-Era. Even though the overall OHC estimates agree during most of the 1970s and 1980s, the spatial patterns do not show a lot of resemblance on annual timescales.

3.4. Future: Reconstructing New Timesteps

In a final analysis step, the utility of the neural network in the reconstruction of data from more recent time period is examined. The same trained neural network is given observations from October 2020 until October 2021. As the training data of the neural network only runs until October 2020, it has never see the observational patterns as well as data assimilations from that period. Therefore, this part is a test of the neural network's ability to replace the EnKF assimilations in its application as a reanalysis for present day timesteps.

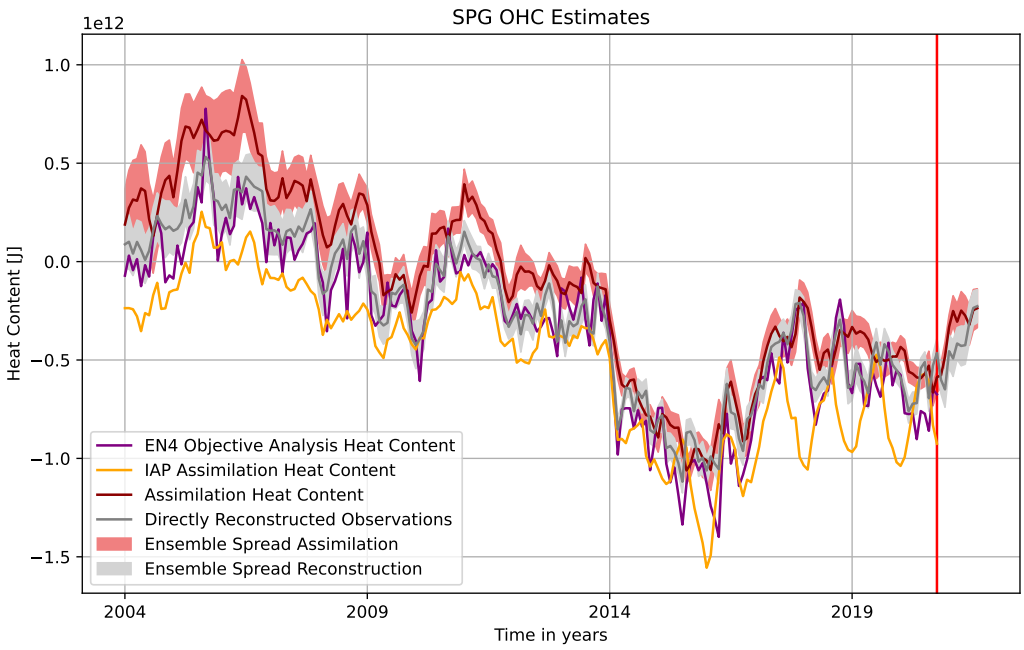

Figure 3.20.: *Estimates of monthly upper 700 m SPG OHC of one EnKF assimilation's ensemble member (red), the direct neural network reconstruction (grey), the IAP (yellow) and the EN4 (purple) during the Argo-Era. In addition to the training period 2004 – 2020, which is marked by the red line, a year of additional more recent observations has been given as input to the neural network.*

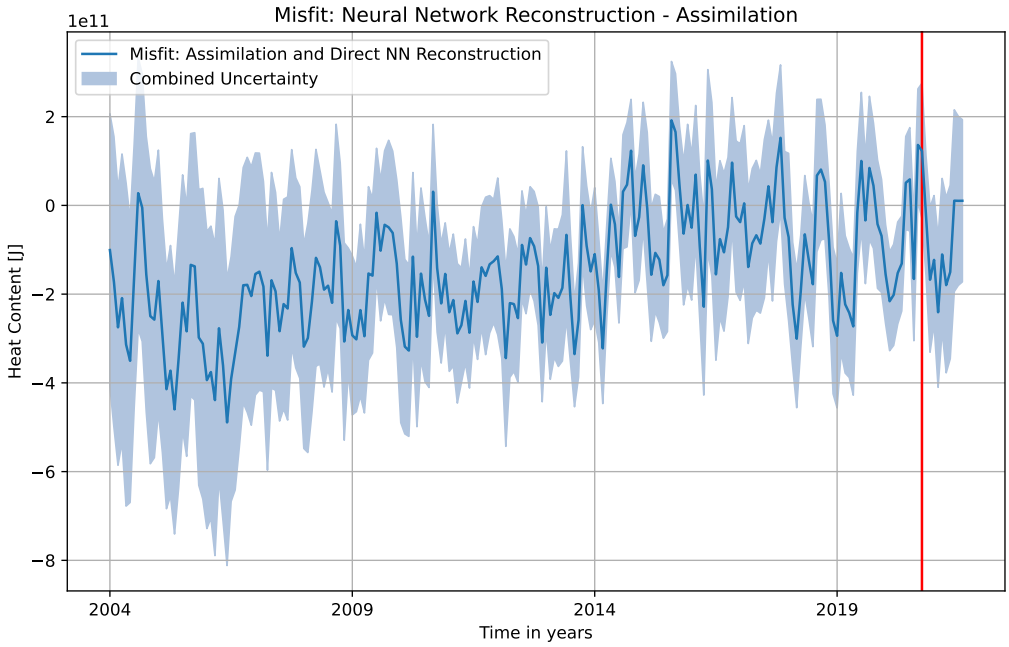

Figure 3.21.: *Misfit between EnKF assimilation NA upper 700m OHC and the neural network's reconstruction during pre-Argo and Argo-Era.*

The resulting reconstruction depicted in figure 3.20 shows a correlation of 0.97 over the newly reconstructed time steps. The start of the new observation input in October 2020 is marked by the red line. It can be seen that nearly all monthly values lie within the uncertainty of the assimilation's ensemble spread. More importantly, the agreement between both timeseries does not deteriorate once the training period ends. The neural network's estimate even lies closer to the assimilation than previously. The misfit between assimilation and neural network reconstruction shows these developments even clearer. Figure 3.21 shows that the differences between both OHC estimates remain very low even after the end of the training period. Again, the negative bias of the direct reconstruction can be explained by the assimilation's bias at the grid points of observations. The comparison to the En4 objective analysis in particular also once again shows, how well the other estimates agree with the direct reconstruction. The higher agreement and correlation compared to the assimilation underline the uncertainty of the integration of observational data into the assimilation, which results in its depicted warm bias.

3.5. Neural Network Uncertainty

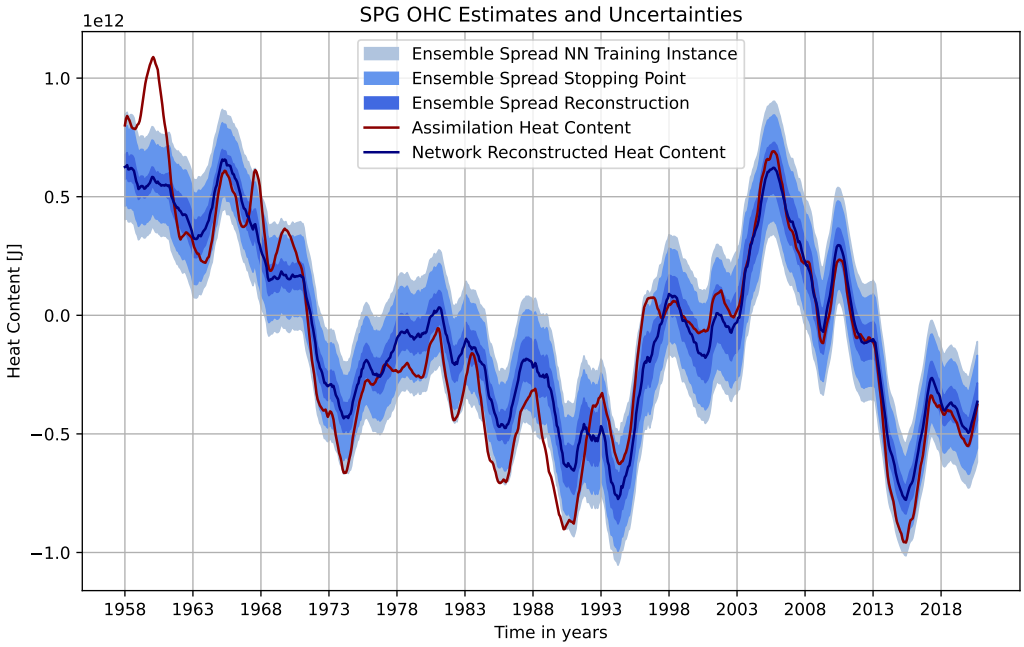

Figure 3.22.: *Estimates of machine learning uncertainty surrounding the reconstruction of SPG OHC (blue line) and compared to the assimilation (red line). Shown are the uncertainty stemming from the reconstruction of the 16 assimilation members (darkest blue shade), the uncertainty stemming from 16 different stopping points of the training process (lighter blue shade) and the uncertainty stemming from the 5 completely new training instances of the neural network (lightest blue).*

The analyses of the Argo and pre-Argo-Era reconstructions of the neural network have already described several sources of uncertainty in the use of neural networks. The first and most obvious source of uncertainty consists of the amount of points of observations. The neural

network does come much closer to reproducing the assimilation during the pre-Argo-Era if the assimilation is masked with Argo-Era masks (figure 3.10). As discussed in section 3.2, this difference cannot be interpreted as a fault of the neural network but rather as a measure of uncertainty within the neural network's training data during times with low observational density. However, other sources of uncertainty, which stem directly from the neural network setup, have not been discussed yet.

The first source of uncertainty inherent to the neural network is the length of its training. Even though there is a validation error considered in the choice of stopping point, it is only a rough estimation of the most accurate version of the neural network. This point can only be evaluated every 5000 iterations, when the model state is saved during the training process. Evaluating the network more often during the training process prolongs the training significantly and would thus not be sensible under a efficiency standpoint. Nonetheless, the remaining uncertainty stemming from the final iteration can be estimated. The uncertainty range around the OHC timeseries of the neural network reconstruction of the validation assimilation ensemble member in figure 3.22 includes additional terms. A 16 member machine learning ensemble is created by saving the model at 16 different points around 550000 iterations with each member 500 iterations apart from the next. The standard deviation of the resulting reconstructions is then shown as the second part of the reconstruction's uncertainty range. As previously, the first part depicts the ensemble spread of the 16 reconstructed assimilation members.

The neural network's second source of uncertainty is the training process itself. The utilized training data is randomly sampled at each training iteration in order to prevent any biases stemming from its selection. Due to this random sampling the neural network develops slightly different at each training iteration. Therefore, the network could move into different local loss minima and consequentially into different network states. In order to estimate this uncertainty, the training process has been repeated five times. The resulting standard deviation of the five reconstructions of this ensemble member is depicted as the third part of the uncertainty range shown in figure 3.22.

3.6. Time and Computational Resources

One of the main reasons to utilize machine learning in the context of data assimilation are the differences in computational resources used. In this section, the actual time and computational resources of the utilized partial convolutional U-net are evaluated and put into the context of the EnKF assimilation. The computational resources needed for all processes are measured in node hours at the high performance computing (HPC) system *Levante* run by the German Climate Computing Center (DKRZ). The neural network is run on NVIDIA's A100 gpu nodes, while the assimilation's node hours are assumed to run on AMD EPYC CPUs (Milan). Further information on the Levante HPC system as well as detailed documentation of its technical setup can be viewed under `https://docs.dkrz.de`.

The computational resources needed by both approaches have to be put into the context of different applications. Table 3.1 summarizes the computation times required by the neural network and the EnKF assimilation for different tasks or scenarios. The computational requirements are once estimated in node hours (NH) to approximate the overall computational costs. Addition-

ally, the overall computation time (OCT) is estimated in hours, which allows for the running of parallel processes and thus approximates the time between start of all computations and the availability of the results. This estimate of the OCT assumes optimal node availability, which in the case of the neural network means 20 gpu nodes, while the assimilation requires 128 cpu nodes to run simultaneously. All resources required by the EnKF assimilations are estimated by (Brune, pers. comm.). The estimates given for the machine learning approach do not include the generation of training data. Thus, it has to be kept in mind that the neural network is dependent on the availability of assimilation or other similar model data in order to perform any reconstruction task. All estimates are rough approximations aiming at showcasing only the dimension of the time required.

Table 3.1.: *Comparison: Computational costs*

Application	NH ML	NH EnKF	OCT ML	OCT EnKF
Assimilation of a new year of observations	0.5	2	0.1	0.25
Generation of an entire 16 member ensemble (1958 – 2022)	400	2048	40	64
Generation of a new ensemble member	2	2176	0.3	68
Adjustment for changes in observations	2	2048	0.3	64

One of the largest differences in computational requirements between both approaches is the adjustment of the reconstruction with changes in observational data. These changes could include new bias corrections for measurements or the inclusion additional and elimination of existing observations. In the case of the machine learning, the trained network would just be required to reconstruct the adjusted observational data. Training the neural network again would be unnecessary, because small differences between training and testing datasets are to expected for example due to assimilation biases. Thus, an adjustment in observations would not significantly impact the performance of an already trained network. Therefore, it would perform such an adjustment very efficiently. The assimilation however would be required to generate an entirely new ensemble, which leads to significantly larger computational requirements. Another aspect in which both approaches differ significantly is the generation of new ensemble members. Due to the fact that EnKF assimilations work only as an ensemble approach, the generation of one new ensemble member requires the recalculation of the entire ensemble.

4. Discussion

The results produced in this thesis, the applied methods and the executed course of action are all based on assumptions. These assumptions and the neural network's limitations have to be discussed first before any final conclusions can be drawn in regard to its performance and potential. Additionally, it can be examined in how far these limitations are specific to this work and can be worked on in further research outside of this thesis.

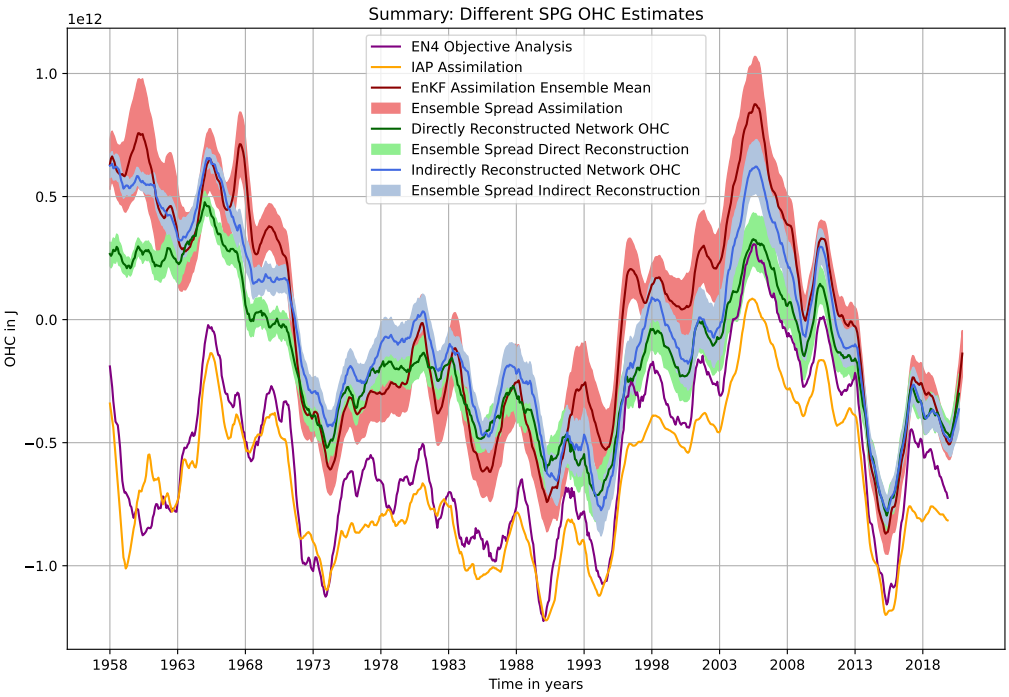

Figure 4.1.: *Comparison of all utilized SPG OHC estimates: EnKF assimilation ensemble mean with its spread (red), the indirect neural network reconstructions with their ensemble spread (blue), the direct neural network reconstructions with their ensemble spread (green) and the IAP (orange) and EN4 (purple) SPG OHC timeseries.*

The neural networks are only trained on temperature profiles produced by the EnKF assimilations. Therefore they are the neural network's main point of reference and the basis for its evaluation. Thus, the entire analysis bases on the assumption that these assimilations provide a close to accurate description of the physical reality during the Argo-Era. In the training process, the neural network learns to reproduce the assimilations spatial patterns and overall trends. In turn, that means that the neural network is limited by its training data's uncertainties and errors. While it can transfer these patterns to times with lower observational densities, it can never outperform but at most reproduce the assimilation in times with the equal or more observations compared to the training period. Additionally, it also reproduces any uncertainties or

errors that are contained in its training data as well as the spatial patterns that are produced by the assimilations specific set of parameters and inputs. Figure 4.1 shows that estimates of SPG OHC can differ significantly and that the neural network is clearly biased towards reproducing its training data. A way to counter this bias and at the same time better estimate the uncertainties of the assimilation during the training period could be the implementation of multi model training data for the neural network. By introducing the training data from different but comparable assimilations, it could help identifying uncertainties due to model biases and uncertainties.

In addition to the limitation due to its training data, the neural network reconstructions have also shown do be dependent on the time period from which their training data is taken. This work is based on the assumption that the physical patterns learned by the neural network during the Argo-Era are directly applicable to the reconstruction during the pre-Argo-Era. However, natural variability with periods longer than the training time as well as changes due to anthropogenic influence can impact the reconstructed variables significantly. Thus, it has to be further examined if a training period of 17 years (the Argo-Era up to 2020) is long enough for the neural network to capture the impact of different modes of low frequency natural variability and anthropogenic influence. Similarly, it also remains to be examined how far into the future the neural networks reconstructions can be applied with a training period of 2004 – 2020. Therefore, the training period has to be extended at some point when reconstructing newer timesteps, especially with rapid changes in regional patterns, such as Arctic heat waves (Barkhordarian et al., 2022). Furthermore, when examining how far into the past the neural network reconstructions are valid, the question also arises how many points of observation are required by the neural network. The results of this work show that up to 1958 the neural network is able to reproduce the assimilation. However, before 1958 the observational density in the NA decreases even further. Which amount of observations is necessary for the neural network in order to still produce significant results, should be examined in further research.

Another key limitation of this work, which has already been indicated, is the utility of the reconstruction created in this thesis. All analysis steps have focused on directly comparing the different OHC estimates without examining its further uses. In current research, assimilations are mostly used to initialize climate hindcasts (Brune and Baehr, 2020). If the reconstructions produced by partial convolutional U-nets are also able to provide hindcasts with meaningful initialization has yet to be examined. Therefore, replacing or even only assisting EnKF assimilations with neural network reconstructions can only be seriously considered after its utility for initialization has been proven.

On a more technical level, an analysis of the uncertainty stemming from the machine learning shows that even the remaining differences between the assimilation and the reconstructions can be explained. Except for the first 5 – 10 years of adjustment period, uncertainties in the training process of the neural network can account for the differences between the OHC estimates. However, these uncertainties are also based on the assumption that the calculated metrics accurately capture the neural network's ability to reproduce the assimilation. The progression of the training loss curves have already shown that loss parameters can have significant shortcomings in measuring the neural network's performance. Therefore, it can be assumed that for example the

choice of stopping point as a result of the examination of the validation loss still does not capture the full extend of the contained uncertainty. This uncertainty could be limited by further hyperparameter training of the loss parameters or the learning rate. But not only the hyperparameter training could be improved. The neural network could also be assisted by the implementation of other additional functions, such as long-short-term-memory (LSTM) or attention modules could be provided. Especially LSTM modules could prove to be helpful by providing information in the time dimension to the reconstruction. Alternatively, the neural network could be improved by adding additional physical information, such as additional variables in the different input channels. In the case of the reconstruction of temperature profiles, the addition of variables, such as salinity or 2m-air temperature could be considered. Another possible modification of the neural network could be the implementation of vertical connections between the depth layers. By feeding all depth layers into the input channels of one larger neural network, vertical connections could theoretically improve the neural network's reconstruction capabilities. However, a rough version of this neural network has been implemented and did not yield an OHC estimate significantly closer to the assimilation while requiring much higher computational resources. Most likely, the neural network is overburdened with the free input of additional degrees of freedom due to the vertical connections. It is often assumed that additional information results in better neural network performance. However, increasing the dimensionality of the training data while keeping its amount the same gives the neural network less possibility to train every single dimension (Bach, 2017). Thus, it has difficulty reconstructing the single temperature layers without additional training data, which then in turn degrades the overall performance. Nonetheless, an improved version of this approach could increase reconstruction accuracy if at the same time the training data is increased and the degrees of freedom are restricted.

In general, the application of this partial convolutional U-net to the reconstruction of other variables other than temperature profiles could examined. Similarly, the reconstruction of temperature profiles in other regions than the NA could be tested. It would be especially interesting to explore in how far the neural network reconstruction still represents a valid alternative to the EnKF assimilations in regions with higher scarcity of observations.

Finally, all estimates of required computational resources in section 3.6 have been made for just this 128x128 grid of the NA. Thus, it could be examined how the differences in computational requirements for both the neural network and EnKF assimilation change with a higher resolution, larger ensembles or larger areas. This could lead to a better estimate of how much time and computational resources the work with machine learning can save. Furthermore, it could explored if neural networks could push resolution of assimilations beyond the EnKF assimilation's current limitations.

5. Conclusion

This study aims at examining if neural networks, more specifically a partial convolutional U-net, represent a valid alternative to the EnKF data assimilations of NA SPG OHC. Additionally, the utility of this neural network to better estimate the uncertainty of data assimilations is inspected.
The neural network is trained on data assimilations from the well observed Argo-Era (2004 – 2020), which are assumed to possess less uncertainty due to the integration of more observations. Then, this trained neural network is tasked to reconstruct masked assimilations as well as observations first from its training period, then from earlier times (1958 – 2004) and finally from 12 newer monthly time steps. The results from these reconstructions are then examined in regard to their agreement to the assimilation as well as other OHC estimates, more specifically those of the IAP and the EN4 objective analysis. On the one hand, this comparison yields conclusions in regard to the neural networks ability to reproduce and replace the data assimilation of three dimensional ocean variables. On the other hand, it helps to estimate the uncertainty of its training data. As the neural network learns the less uncertain assimilation's spatial patterns from well observed time periods, it is not expected that the neural network reconstruction during the pre-Argo-Era agrees entirely with its training data. Instead, the differences between the indirect reconstruction of masked assimilations and the assimilation itself are taken as indicators for uncertainties when the assimilation's model dynamics are not supported by observational input. The differences between the direct neural network reconstruction of EN4 observations and the assimilation underline the assimilation's uncertainty at the grid points of observations and its effect on the overall OHC estimate.

The results yielded by this approach lead me to the following conclusion:

- Given the close agreement between the neural network reconstruction and the EnKF assimilation I conclude that **partial convolutional U-nets are able to reproduce EnKF assimilations during its training period, before and after**. In all cases significant correlations in its overall OHC estimates as well as its spatial patterns are found. Even monthly variations in time steps outside of its training data are reproduced by the neural network when directly reconstructing observations from the Argo-Era onwards. Thus, partial convolutional U-nets provide a valid alternative to enhance and reproduce data assimilation of ocean variables. Additionally, the results demonstrate that the neural network reconstructions are also able to replace the EnKF assimilation in their function as a OHC reanalysis for at least a year outside of its training period.

- Furthermore, this study also demonstrates that neural network reconstructions can take a role in better estimating the uncertainty of EnKF assimilations of OHC. The neural network is able to transfer spatial patterns from its well observed training period to ear-

lier time spans. Consequently, the differences between assimilation and network reconstruction at times with lower observational coverage point towards uncertainties in the assimilation's model dynamics when they are not aided by observational input. This ability is highlighted by the neural network's detection of inconsistencies in the assimilation's misrepresentation of the North Atlantic current's North-West corner at the start of the pre-Argo-Era. The results of the direct reconstruction of observations highlight inconsistencies between assimilation and observations at the observed grid points. Therefore, in addition to replacing them, **partial convolutional U-nets are able to detect inconsistencies and estimate uncertainties in data assimilation outside of its ensemble spread, especially in regard to time periods with low observational density**.

- Finally, an analysis of the computational resources used by both the EnKF assimilations and the neural network shows that **neural networks have the potential to significantly speed up parts of the assimilation process**. The largest differences occur, when the neural network's ability to reconstruct independent of another training instance is utilized. The generation of an additional ensemble member or the reconstruction of adjusted observational data show differences in node hours of up to a factor 1000. With growing resolution and larger datasets and ensembles, this difference between the computational requirements is bound to grow even further (Brune, pers. comm.).

With this study, I show that neural networks have the ability to reproduce, extend and even replace assimilations and other mapping methods. Additionally, they are capable of giving insights into the uncertainty of other assimilation techniques, especially for undersampled regions and variables. Therefore, I overall demonstrate that due to their efficiency, flexibility and ability to learn spatial patterns, partial convolutional U-nets can significantly impact the data assimilation of three-dimensional ocean variables.

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7. List of Acronyms

OHC Ocean Heat Content

NA North Atlantic

SPG Subpolar Gyre

NAO North Atlantic Oscillation

SST Sea Surface Temperature

XBT Expendable Bathythermographs

MBT Mechanical Bathythermographs

KF Kalman Filter

EnKF Ensemble Kalman Filter

ESM Earth System Model

RMSE Root Mean Squared Error

CMIP5 Climate Model Intercomparison Project Phase 5

IAP Institute for Atmospheric Physics

ASBO Arctic Synoptic Basinide Oceanography

GTSPP Global Temperature and Salinity Profile Program

WOD13 Word Ocean Database

ReLU rectified linear unit

MAE Mean Absolute Error

HPC High Performance Computing

DKRZ Deutsches Klimarechenzentrum (German Climate Computing Center)

NH Node Hours

OCT Overall Computation Time

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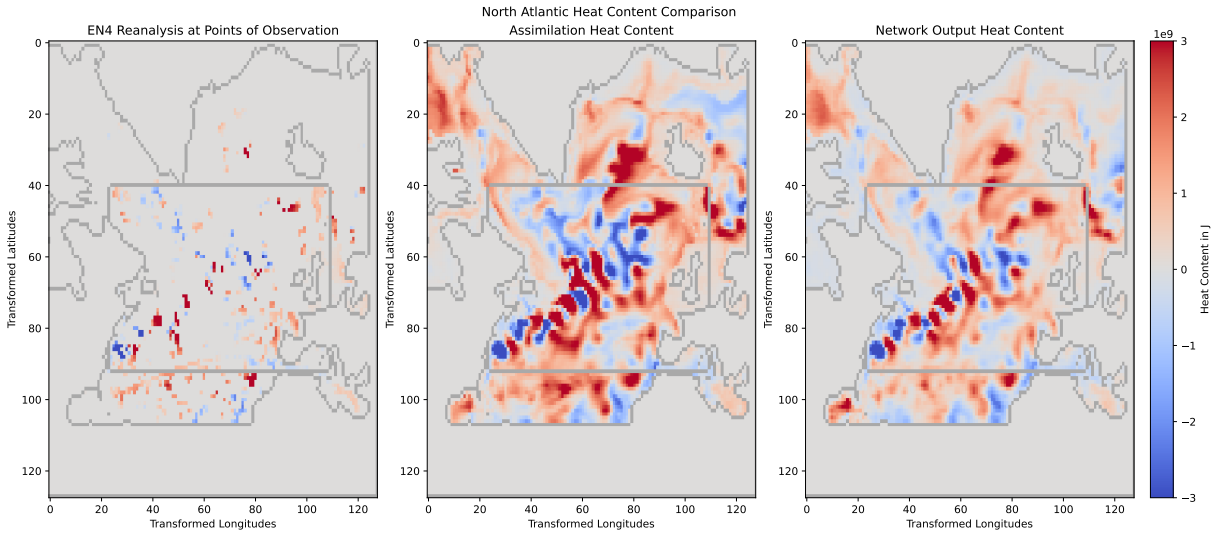


Figure A.1.: *Map of masked EnKF assimilation (left), the assimilation itself (middle) and neural network reconstruction (right) of NA upper 700m anomaly OHC estimate in January 2008. The SPG region that is used for the construction of the timeseries is marked by the grey rectangle. The unusual projection of the North Atlantic is oriented at the assimilation's irregular model grid, which enlarges the SPG region.*

Training Loss: Argo-Mask Training

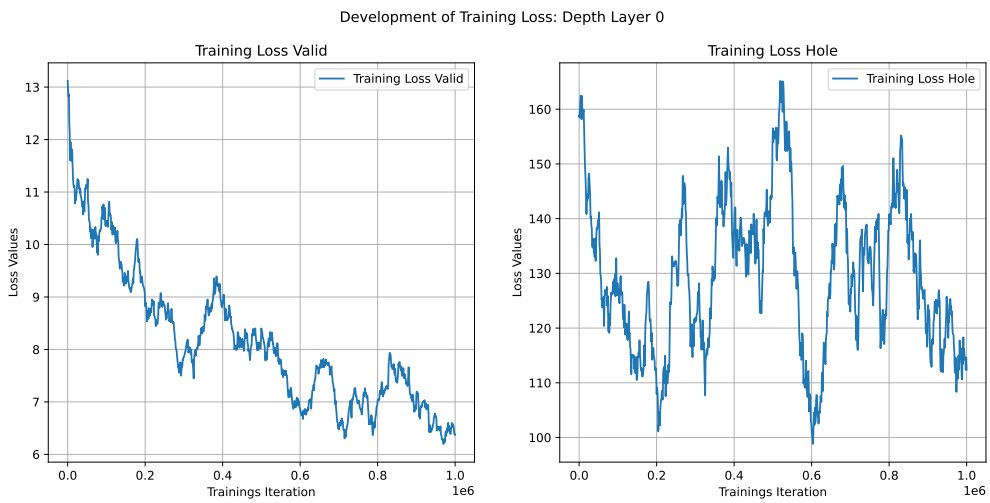


Figure A.2.: *Smoothed hole and valid loss curves for the first depth layer over the 600.000 Training iterations of the Argo-Mask training.*

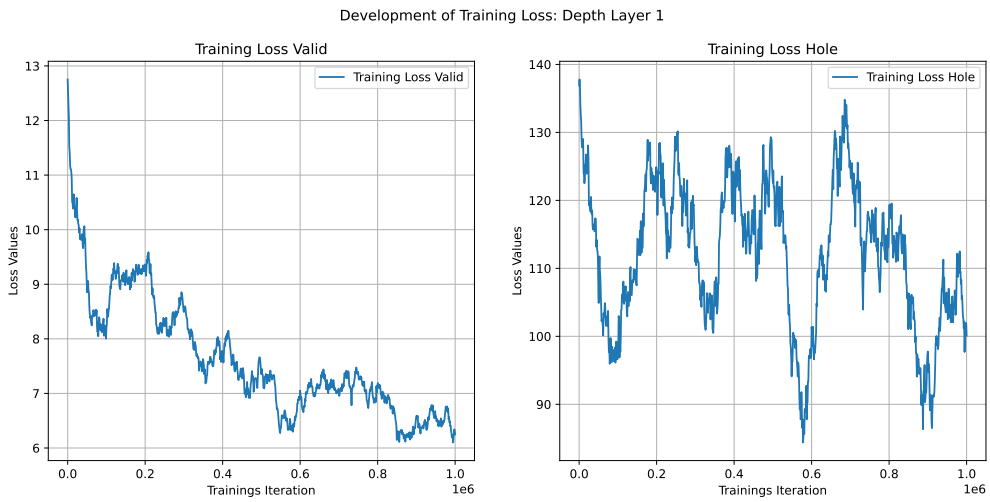


Figure A.3.: *Smoothed hole and valid loss curves for the second depth layer over the 600.000 Training iterations of the Argo-Mask training.*

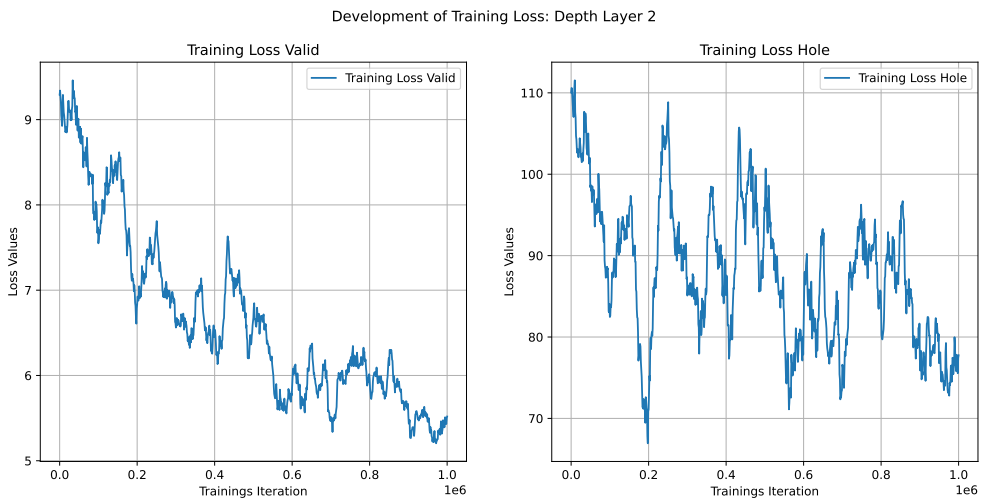


Figure A.4.: *Smoothed hole and valid loss curves for the third depth layer over the 600.000 Training iterations of the Argo-Mask training.*

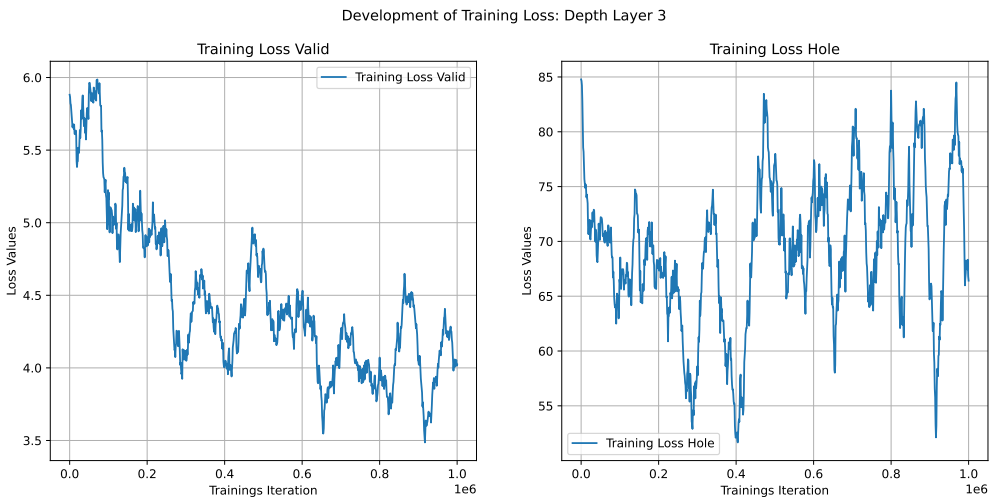


Figure A.5.: *Smoothed hole and valid loss curves for the fourth depth layer over the 600.000 Training iterations of the Argo-Mask training.*

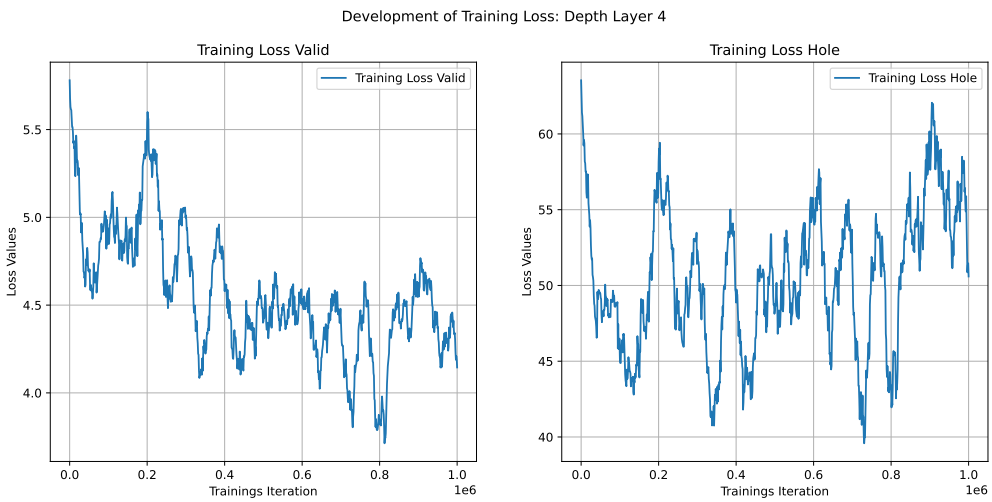


Figure A.6.: *Smoothed hole and valid loss curves for the fifth depth layer over the 600.000 Training iterations of the Argo-Mask training.*

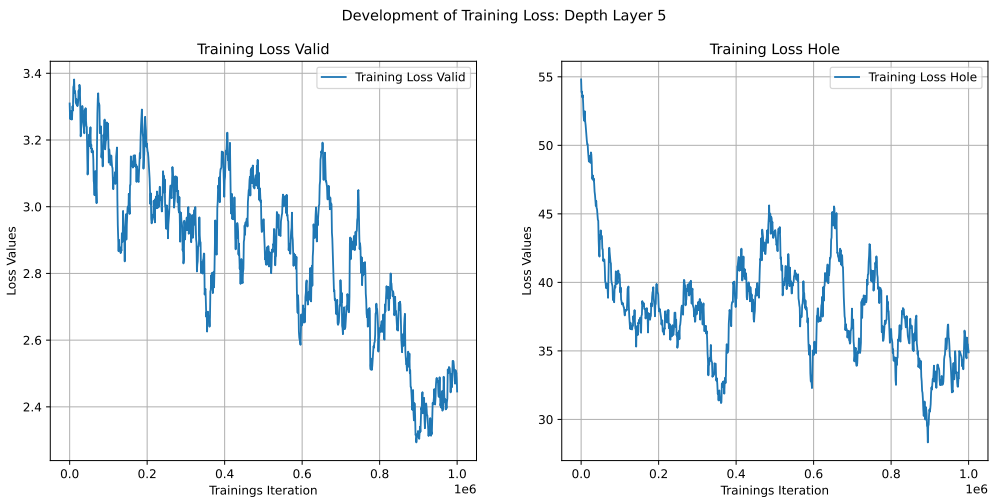
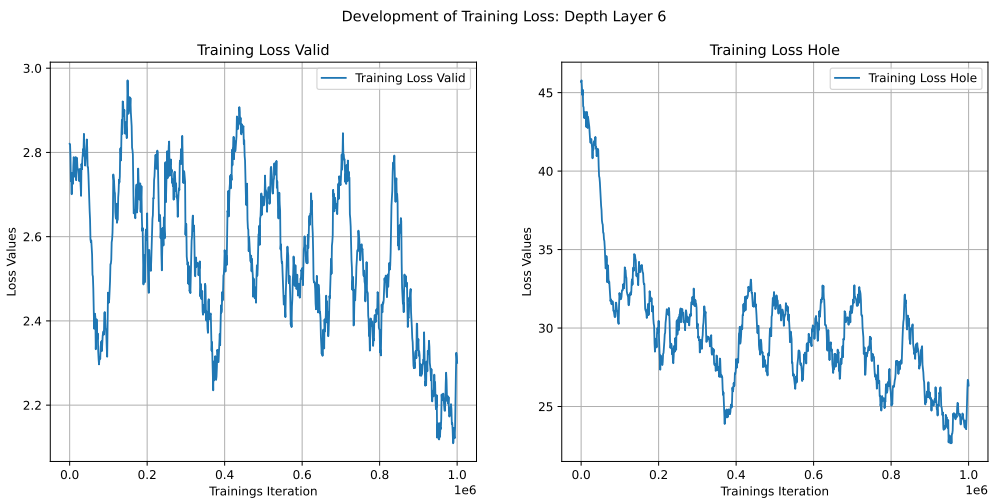


Figure A.7.: *Smoothed hole and valid loss curves for the sixth depth layer over the 600.000 Training iterations of the Argo-Mask training.*



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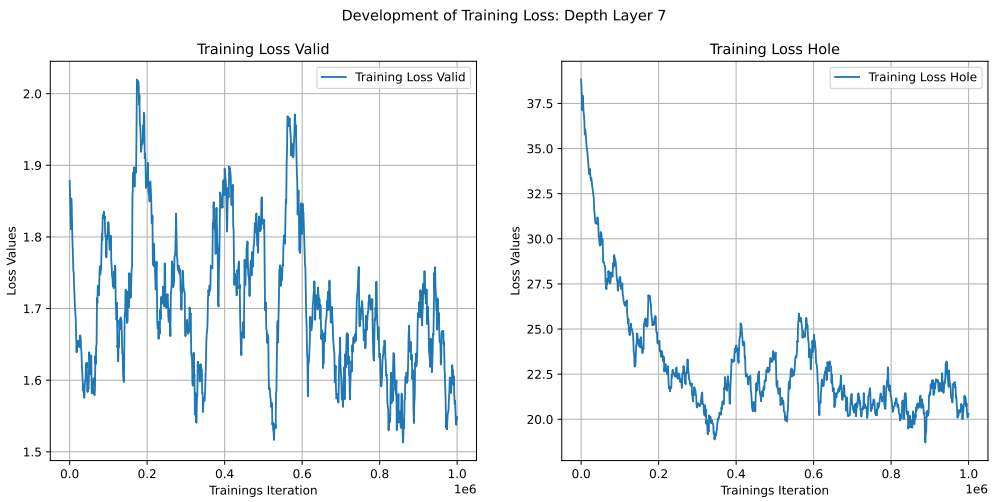


Figure A.9.: *Smoothed hole and valid loss curves for the eighth depth layer over the 600.000 Training iterations of the Argo-Mask training.*

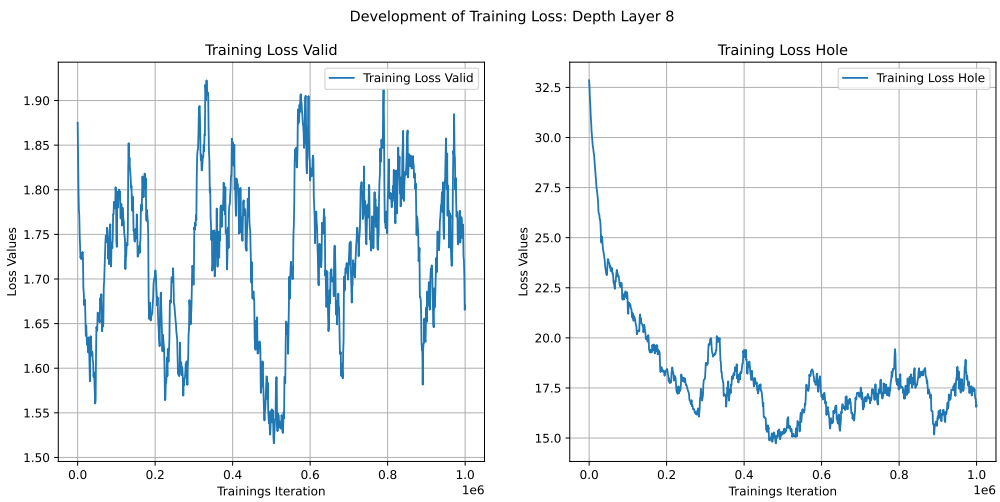


Figure A.10.: *Smoothed hole and valid loss curves for the ninth depth layer over the 600.000 Training iterations of the Argo-Mask training.*

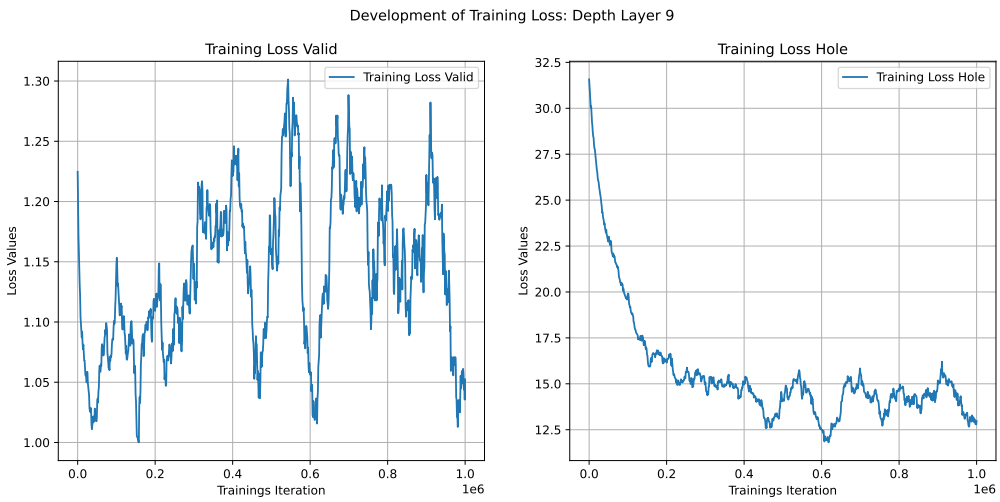


Figure A.11.: *Smoothed hole and valid loss curves for the tenth depth layer over the 600.000 Training iterations of the Argo-Mask training.*

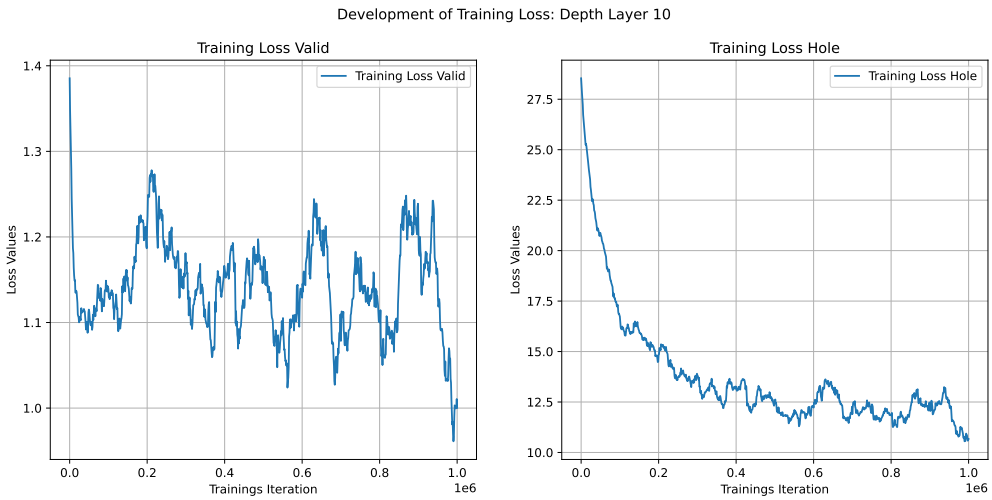


Figure A.12.: *Smoothed hole and valid loss curves for the eleventh depth layer over the 600.000 Training iterations of the Argo-Mask training.*

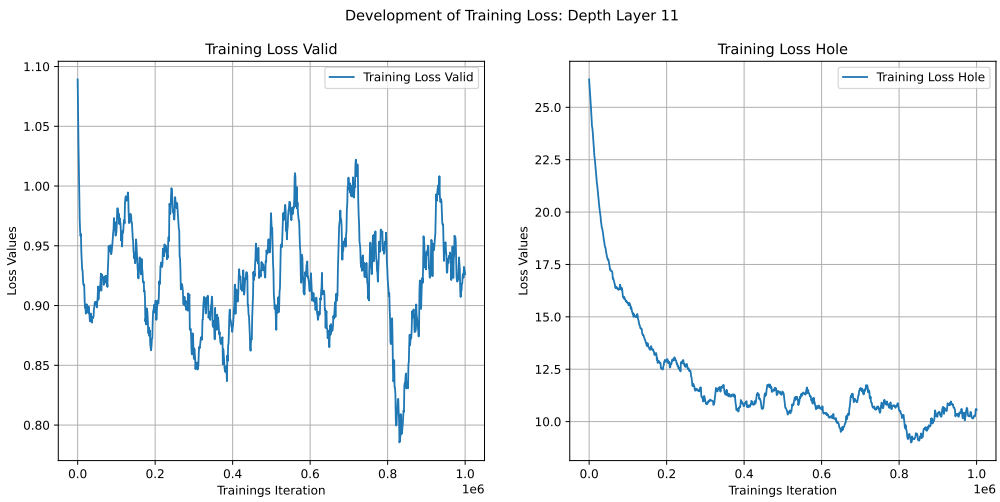


Figure A.13.: *Smoothed hole and valid loss curves for the twelfth depth layer over the 600.000 Training iterations of the Argo-Mask training.*

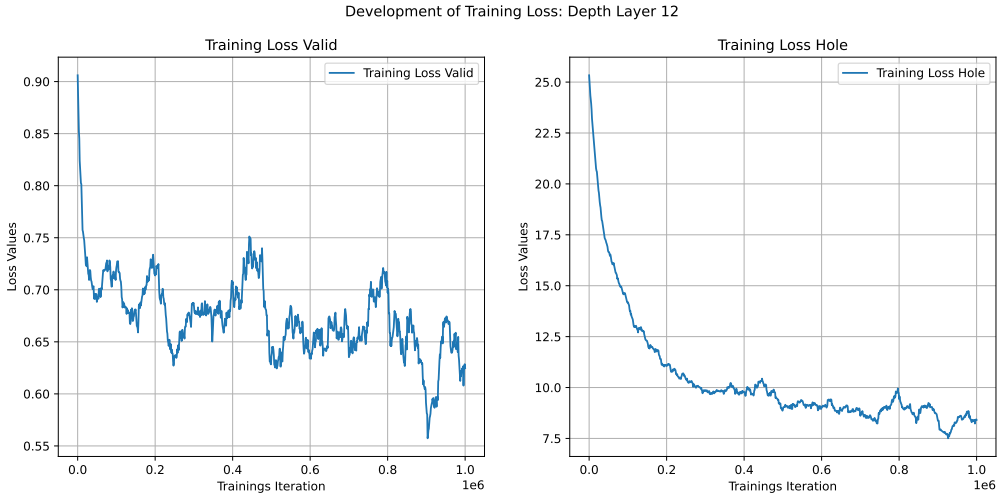


Figure A.14.: *Smoothed hole and valid loss curves for the thirteenth depth layer over the 600.000 Training iterations of the Argo-Mask training.*

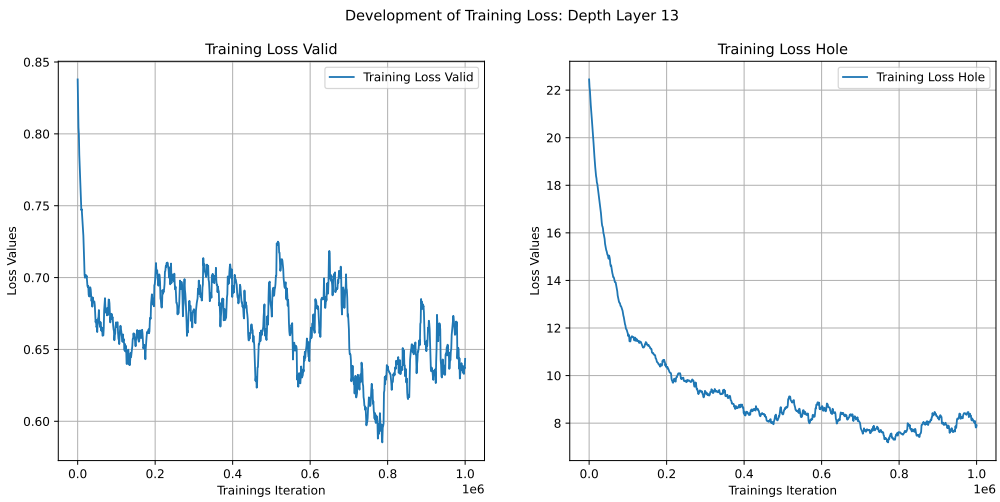
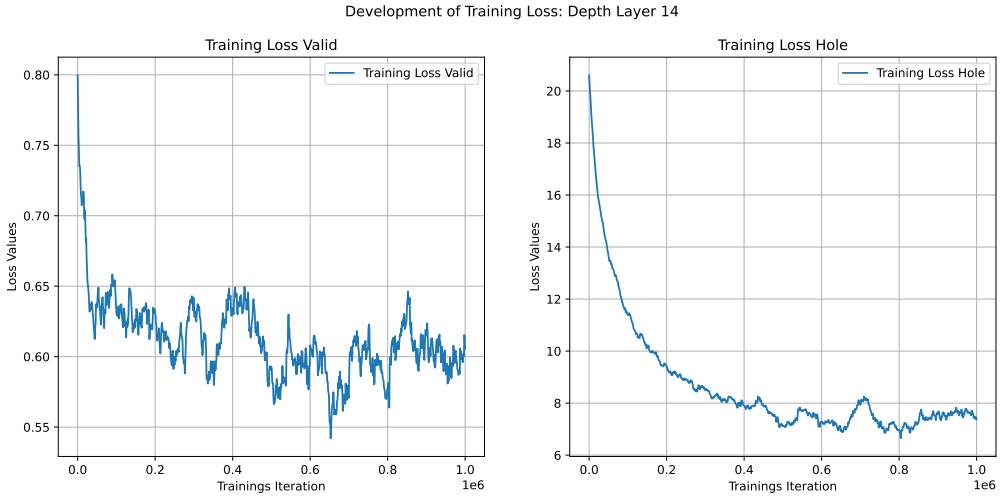


Figure A.15.: *Smoothed hole and valid loss curves for the fourteenth depth layer over the 600.000 Training iterations of the Argo-Mask training.*



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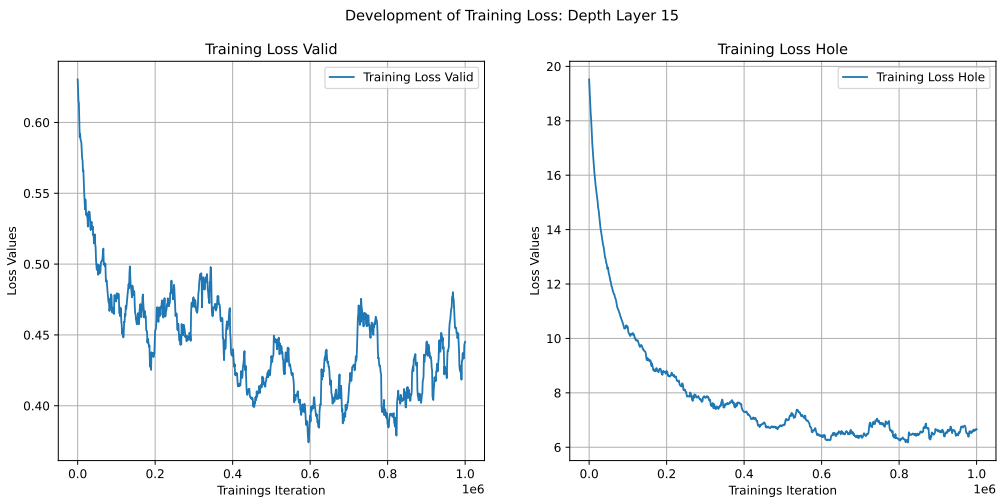

Figure A.17.: *Smoothed hole and valid loss curves for the sixteenth depth layer over the 600.000 Training iterations of the Argo-Mask training.*

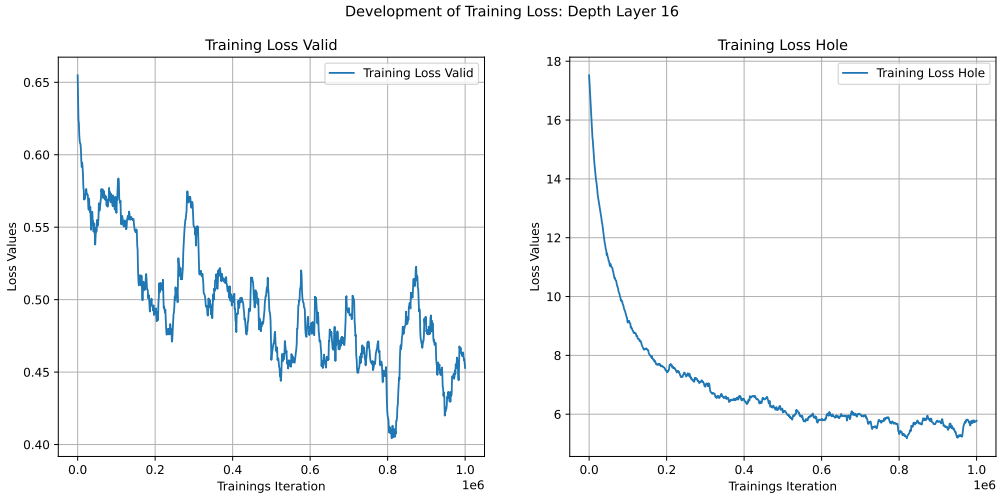

Figure A.18.: *Smoothed hole and valid loss curves for the seventeenth depth layer over the 600.000 Training iterations of the Argo-Mask training.*

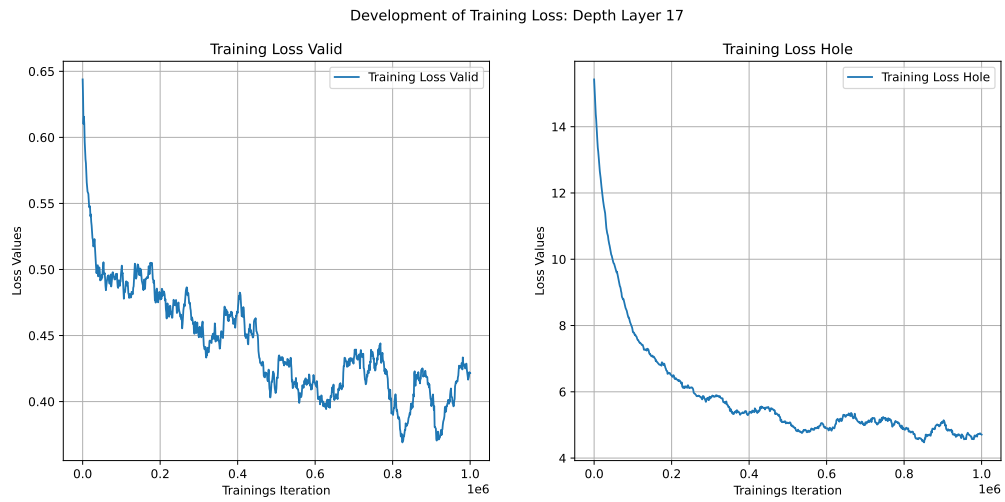

Figure A.19.: *Smoothed hole and valid loss curves for the eighteenth depth layer over the 600.000 Training iterations of the Argo-Mask training.*

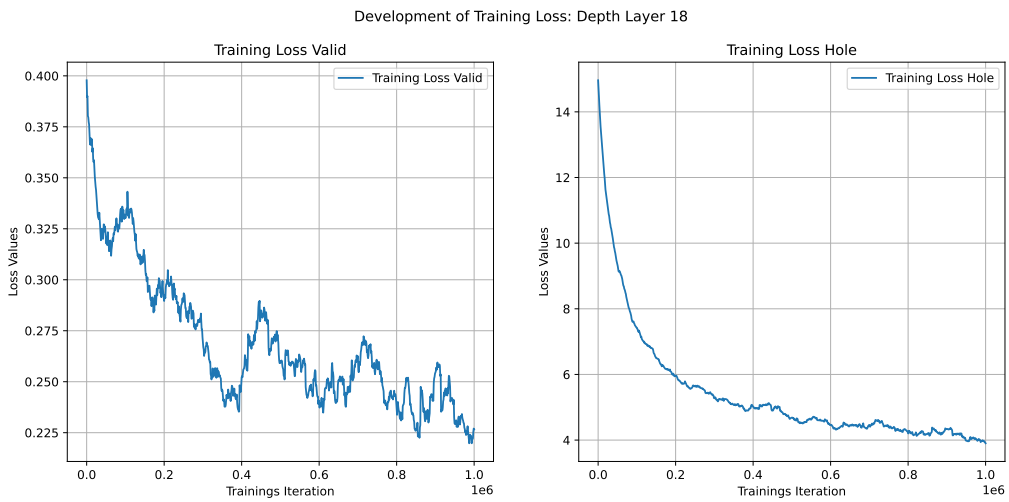

Figure A.20.: *Smoothed hole and valid loss curves for the nineteenth depth layer over the 600.000 Training iterations of the Argo-Mask training.*

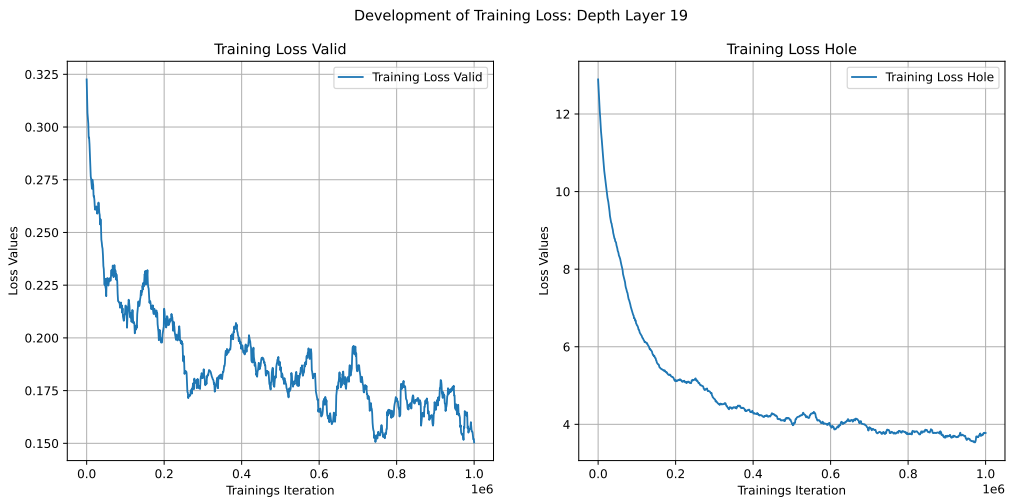

Figure A.21.: *Smoothed hole and valid loss curves for the twentieth depth layer over the 600.000 Training iterations of the Argo-Mask training.*

Training Loss: Full Mask Training

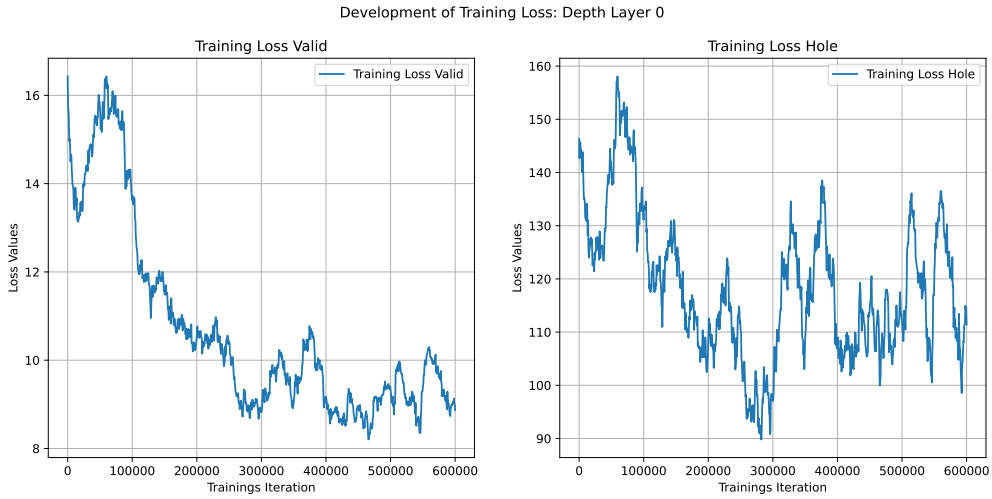

Figure A.22.: *Smoothed hole and valid loss curves for the first depth layer over the 1.000.000 Training iterations of the Full-Mask training.*

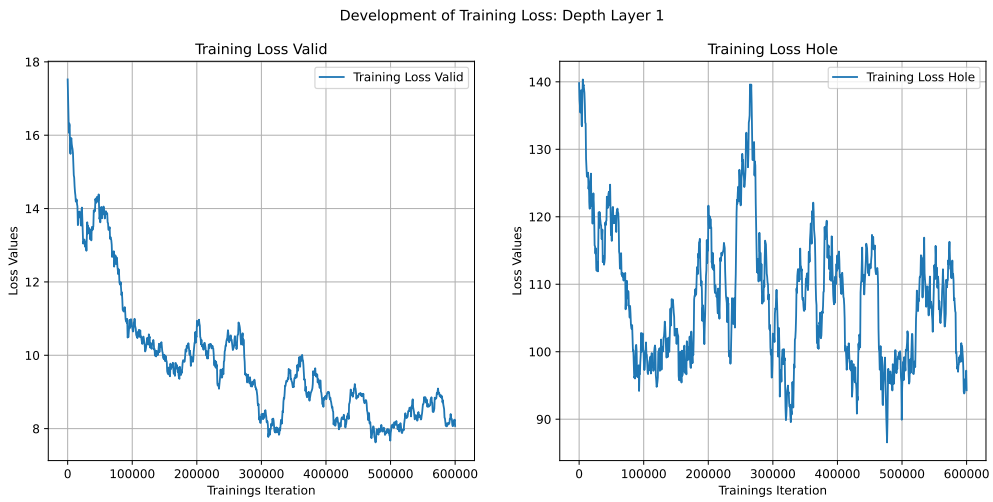

Figure A.23.: *Smoothed hole and valid loss curves for the second depth layer over the 1.000.000 Training iterations of the Full-Mask training.*

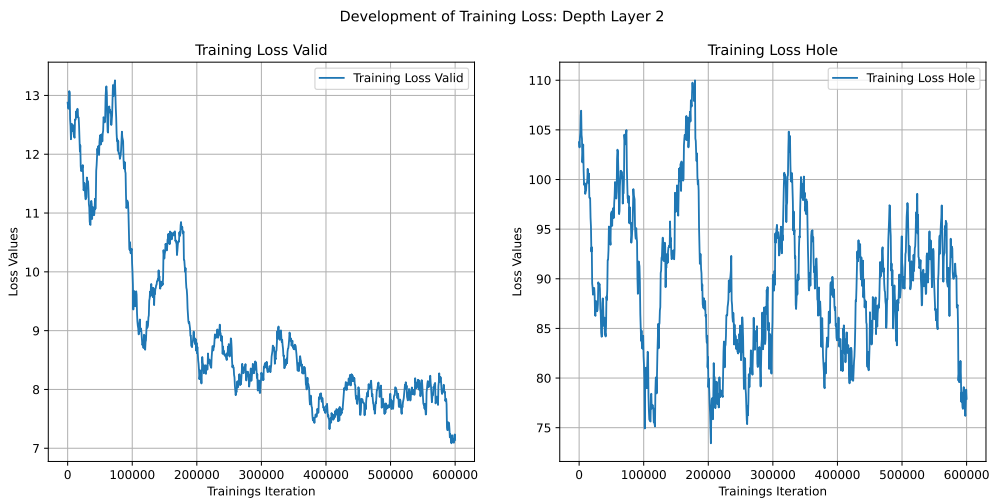

Figure A.24.: *Smoothed hole and valid loss curves for the third depth layer over the 1.000.000 Training iterations of the Full-Mask training.*

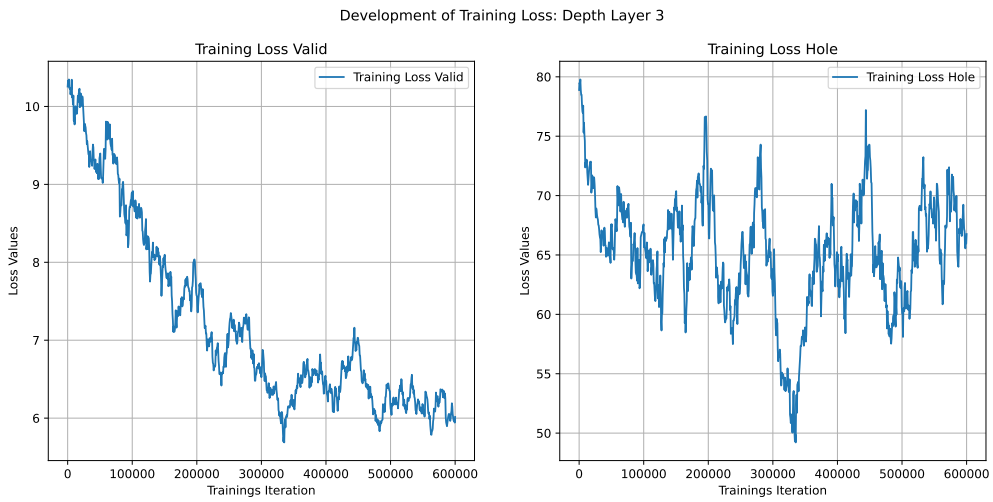

Figure A.25.: *Smoothed hole and valid loss curves for the fourth depth layer over the 1.000.000 Training iterations of the Full-Mask training.*

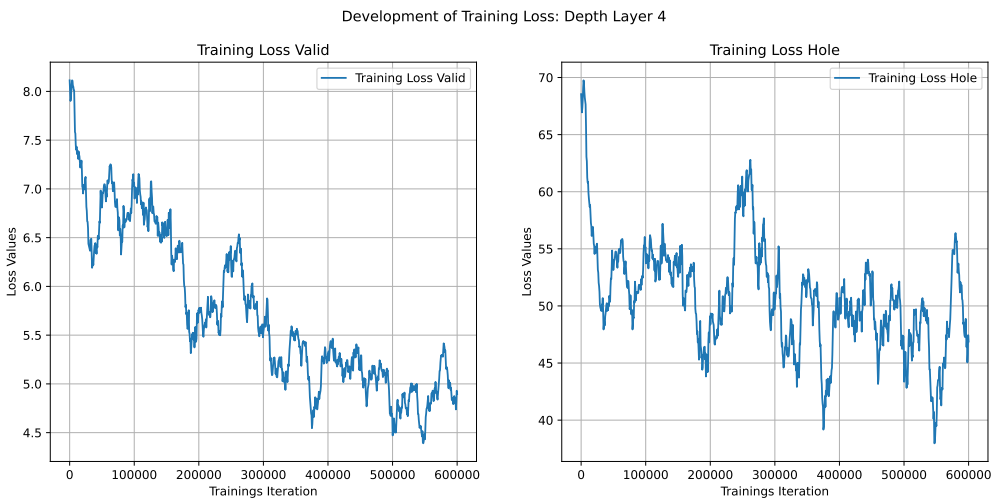

Figure A.26.: *Smoothed hole and valid loss curves for the fifth depth layer over the 1.000.000 Training iterations of the Full-Mask training.*

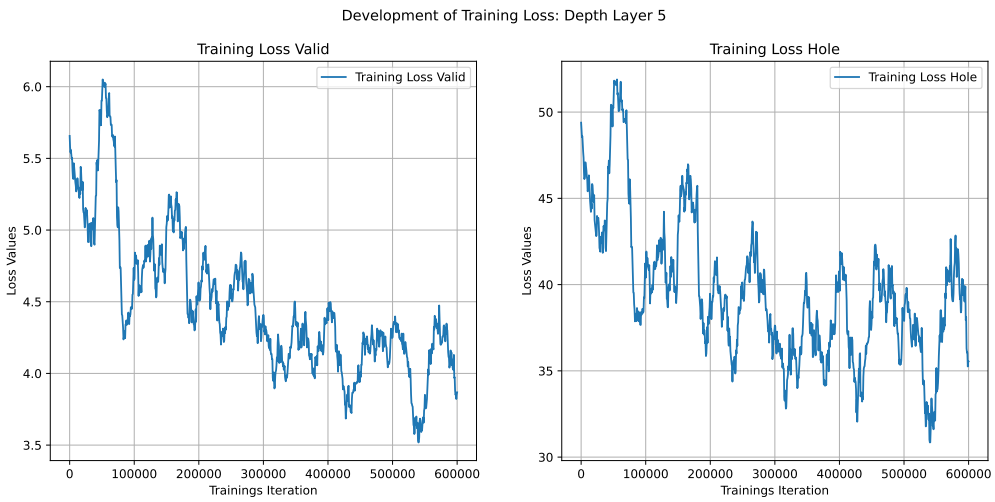

Figure A.27.: *Smoothed hole and valid loss curves for the sixth depth layer over the 1.000.000 Training iterations of the Full-Mask training.*

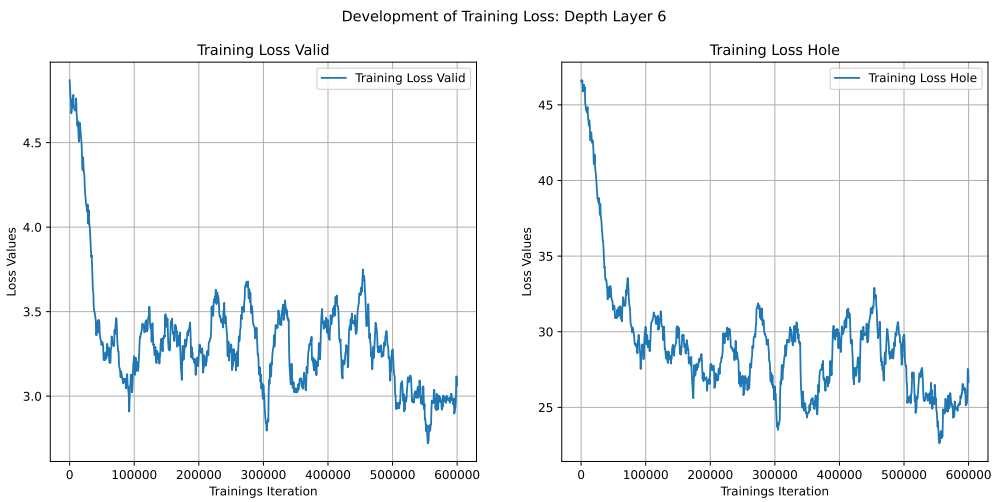

Figure A.28.: *Smoothed hole and valid loss curves for the seventh depth layer over the 1.000.000 Training iterations of the Full-Mask training.*

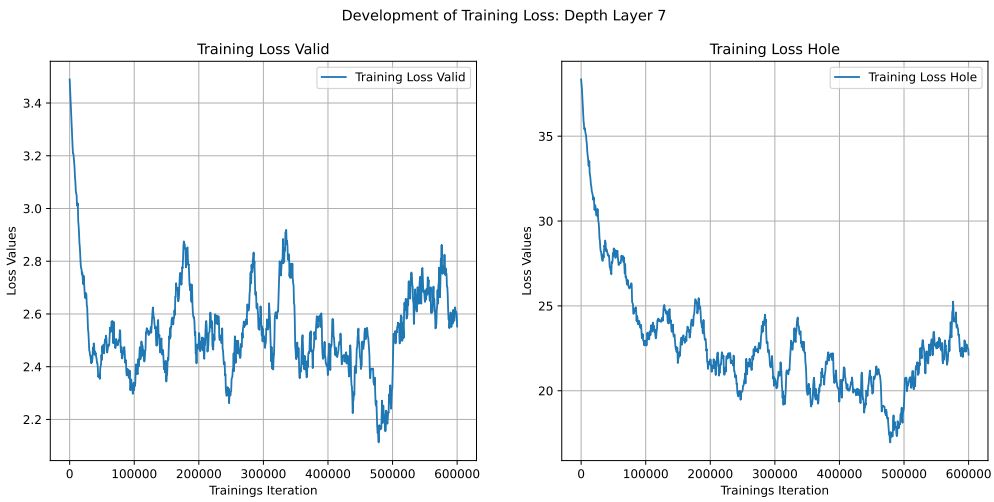

Figure A.29.: *Smoothed hole and valid loss curves for the eighth depth layer over the 1.000.000 Training iterations of the Full-Mask training.*

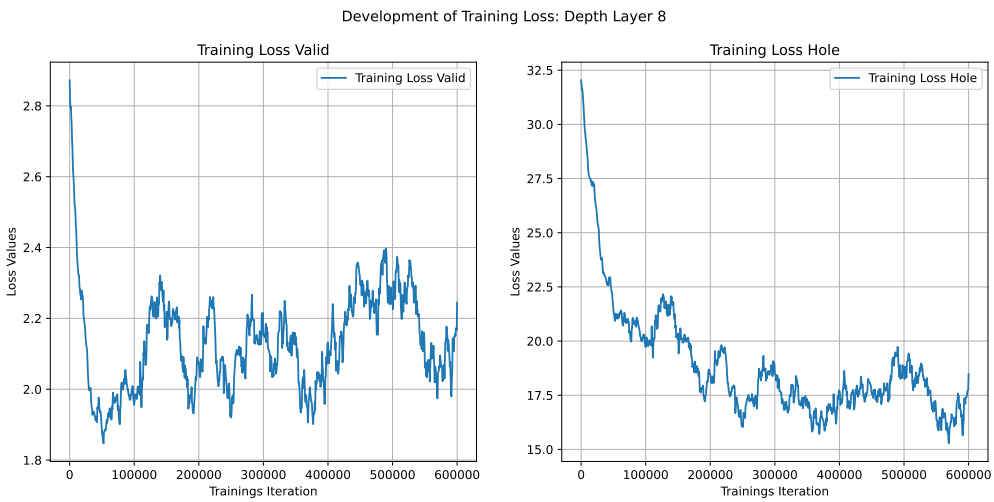

Figure A.30.: *Smoothed hole and valid loss curves for the ninth depth layer over the 1.000.000 Training iterations of the Full-Mask training.*

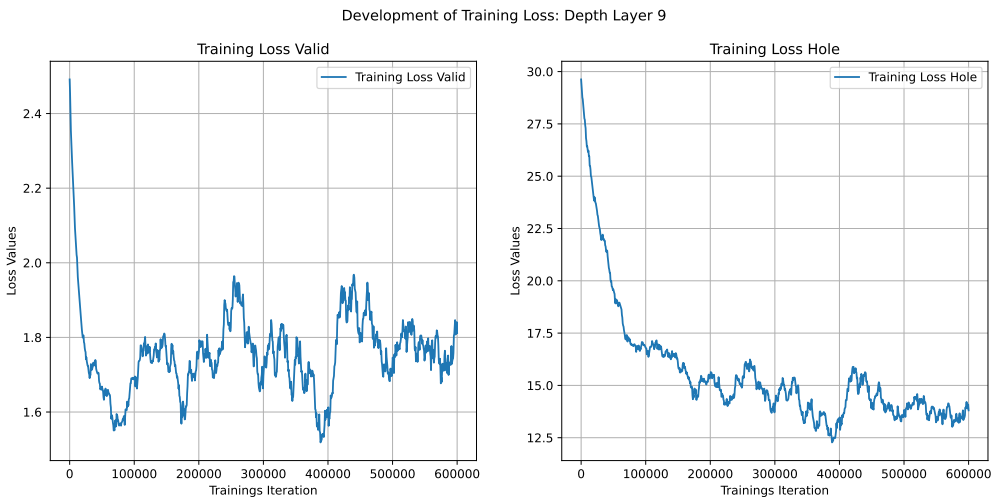

Figure A.31.: *Smoothed hole and valid loss curves for the tenth depth layer over the 1.000.000 Training iterations of the Full-Mask training.*

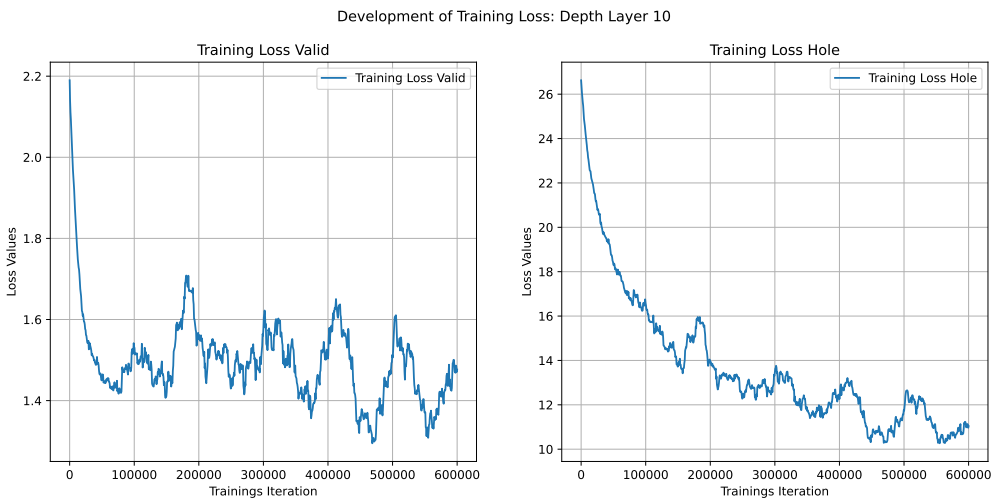

Figure A.32.: *Smoothed hole and valid loss curves for the eleventh depth layer over the 1.000.000 Training iterations of the Full-Mask training.*

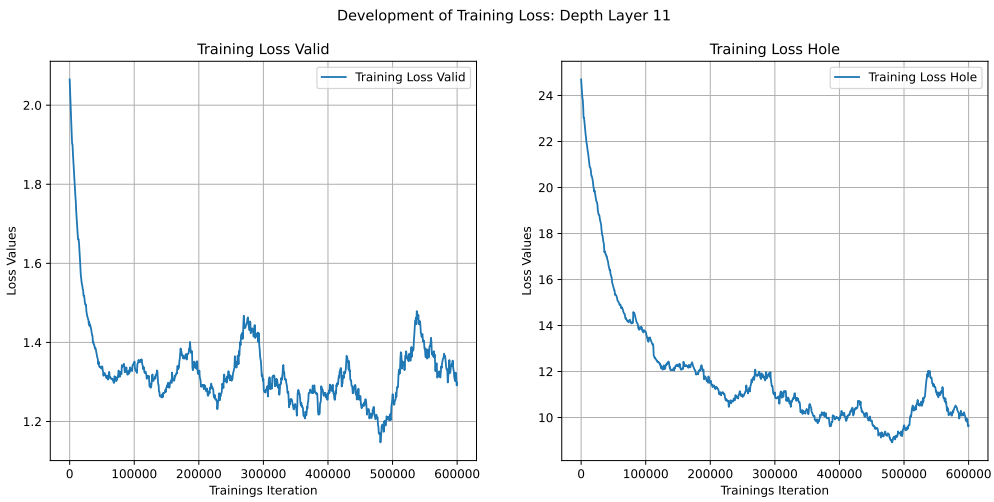

Figure A.33.: *Smoothed hole and valid loss curves for the twelfth depth layer over the 1.000.000 Training iterations of the Full-Mask training.*

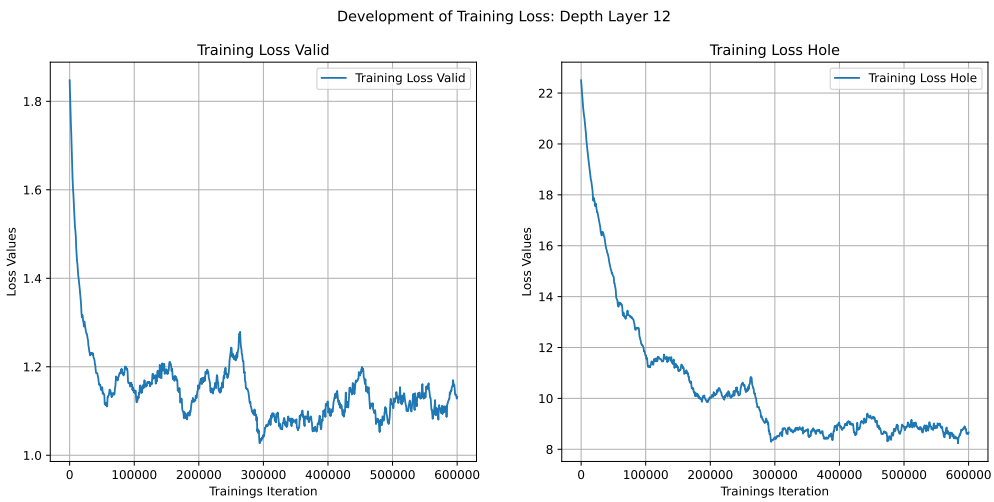

Figure A.34.: *Smoothed hole and valid loss curves for the thirteenth depth layer over the 1.000.000 Training iterations of the Full-Mask training.*

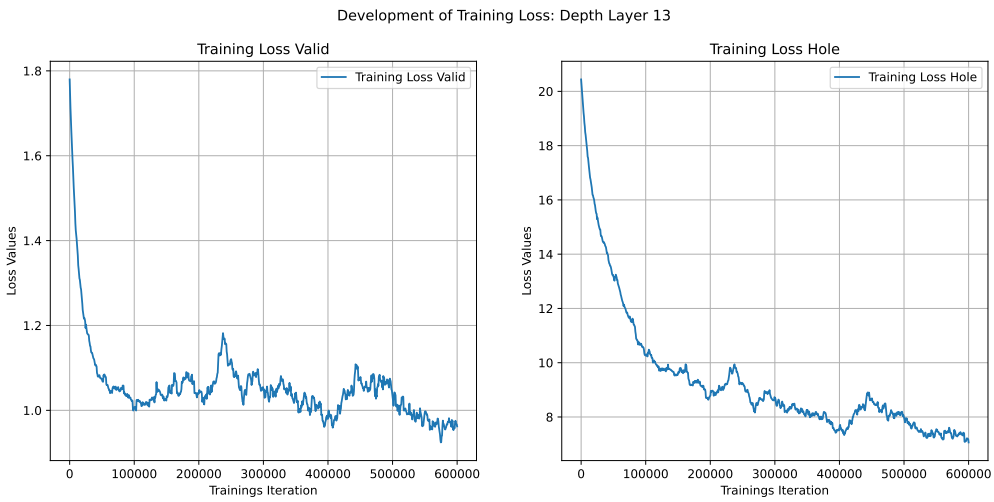

Figure A.35.: *Smoothed hole and valid loss curves for the fourteenth depth layer over the 1.000.000 Training iterations of the Full-Mask training.*

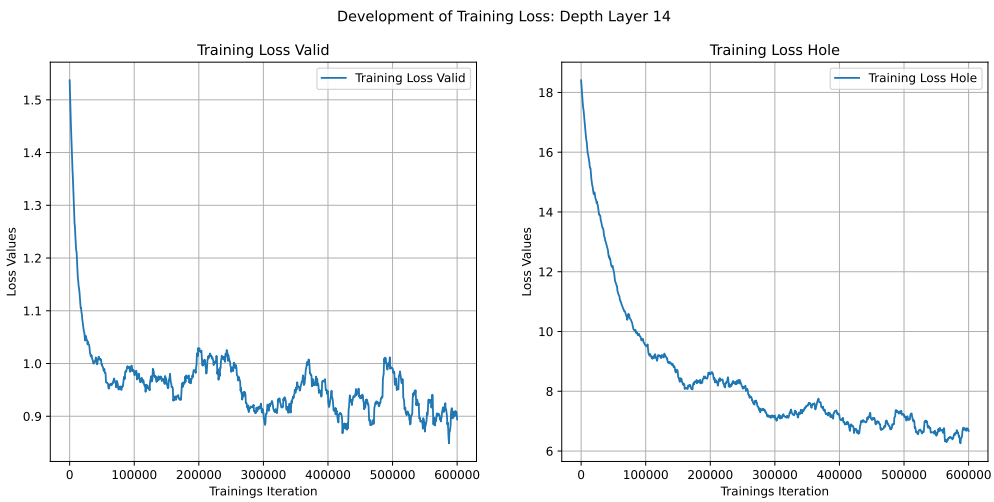

Figure A.36.: *Smoothed hole and valid loss curves for the fifteenth depth layer over the 1.000.000 Training iterations of the Full-Mask training.*

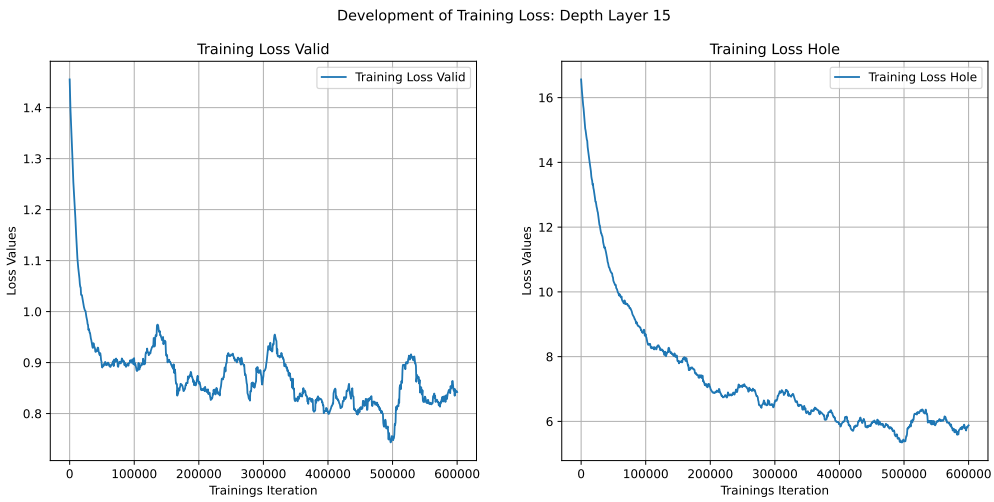

Figure A.37.: *Smoothed hole and valid loss curves for the sixteenth depth layer over the 1.000.000 Training iterations of the Full-Mask training.*

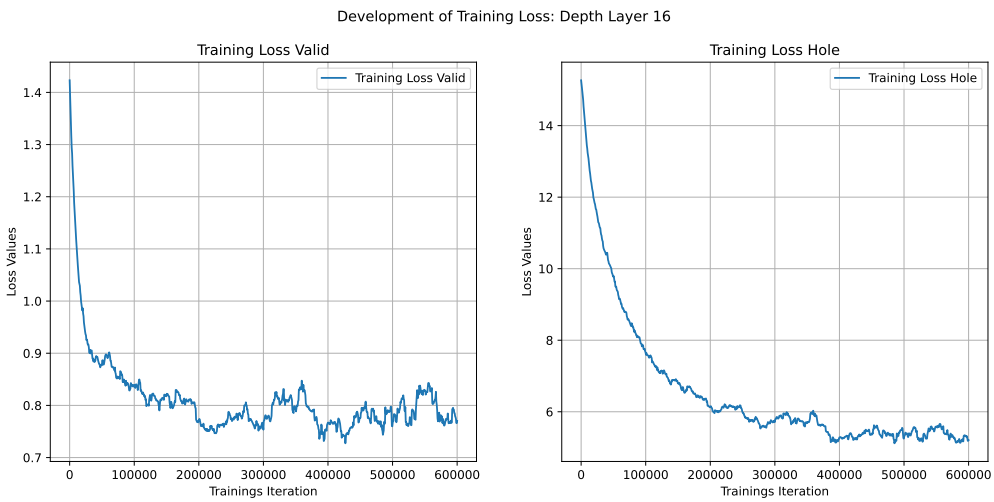

Figure A.38.: *Smoothed hole and valid loss curves for the seventeenth depth layer over the 1.000.000 Training iterations of the Full-Mask training.*

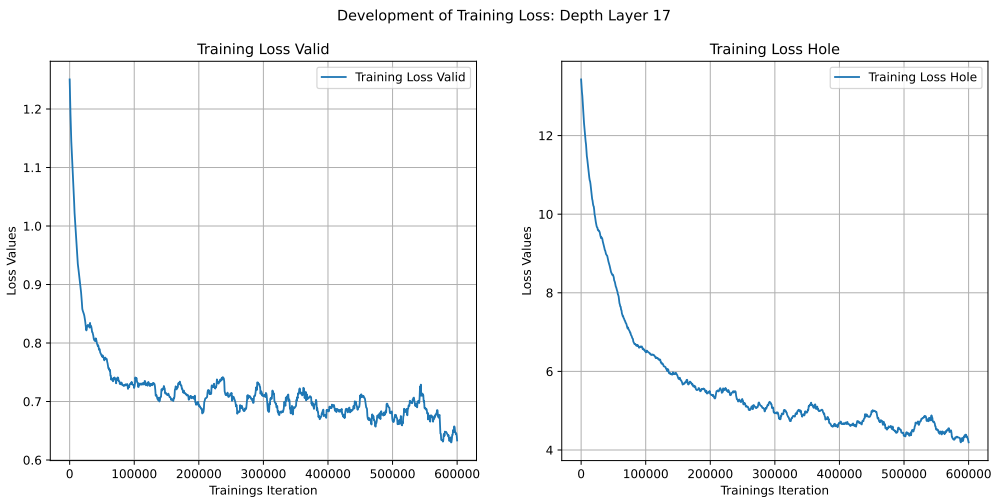

Figure A.39.: *Smoothed hole and valid loss curves for the eighteenth depth layer over the 1.000.000 Training iterations of the Full-Mask training.*

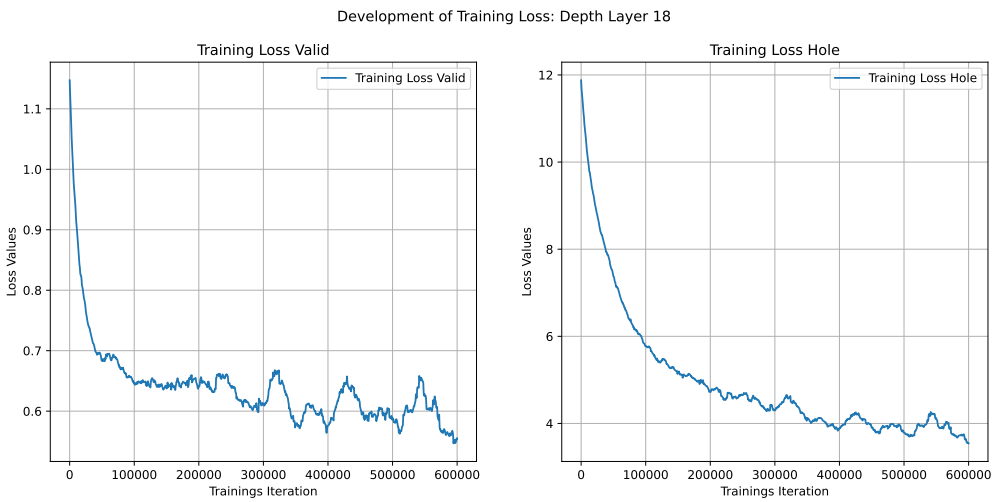

Figure A.40.: *Smoothed hole and valid loss curves for the nineteenth depth layer over the 1.000.000 Training iterations of the Full-Mask training.*

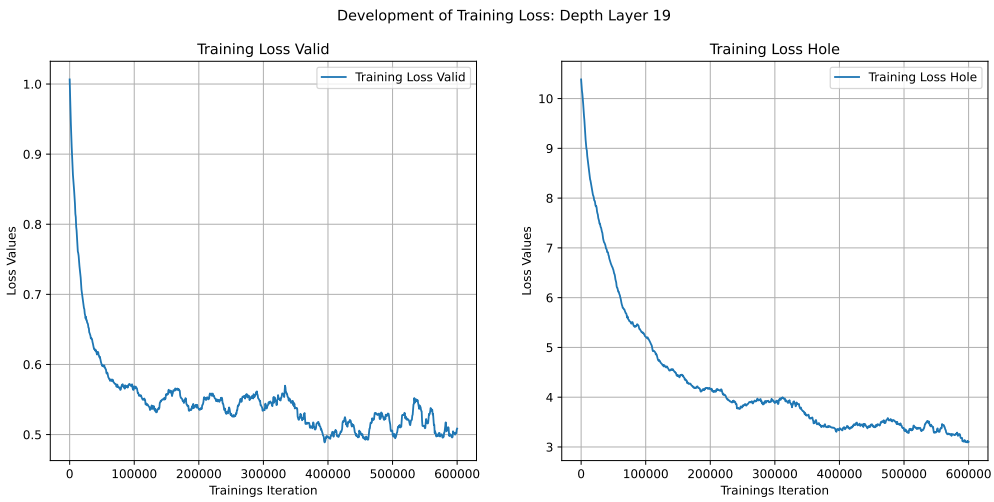

Figure A.41.: *Smoothed hole and valid loss curves for the twentieth depth layer over the 1.000.000 Training iterations of the Full-Mask training.*

Eidesstaatliche Versicherung

Hiermit versichere ich an Eides statt, dass ich diese Arbeit eigenständig verfasst habe. Ich habe keine anderen als die angegebenen Quellen und Hilfsmittel benutzt sowie wörtliche und sinngemäße Zitate kenntlich gemacht. Die Arbeit hat in gleicher oder ähnlicher Form noch keiner Prüfungsbehörde vorgelegen und die eingereichte schriftliche Fassung entspricht der auf dem elektronischen Speichermedium.

13.01.2023